

PROCESSING, STRUCTURE, AND MORPHOLOGY IN POLYMER NANOCOMPOSITES

DURACCIO DONATELLA^{1,2}, CLARA SILVESTRE¹, SOSSIO CIMMINO¹, ANTONELLA MARRA¹, AND MARILENA PEZZUTO¹

¹*Istituto per i Polimeri, Compositi e Biomateriali, Consiglio Nazionale delle Ricerche, Pozzuoli, Italy*

²*Istituto per le Macchine Agricole e Movimento Terra (IMAMOTER) – Consiglio Nazionale delle Ricerche, Torino, Italy*

The term “polymer nanocomposite” (PNC) has evolved, since the first reports in the early 1990s to refer to a multicomponent system, where the major constituent is a polymer or blend thereof and the minor constituent exhibits a length scale below 100 nm [1–3]. The minor constituent is usually an inorganic filler, called nanofiller, nanoload, or, improperly, nanoparticle. The most commonly used are layered silicates (clays), carbon nanotubes (CNTs), and metals and various metal oxides (silica, titania, zirconia, zinc oxide, etc.).

PNCs have generated a significant amount of industrial and academic attention for the past 25 years. No matter the measure (articles, patents, or research and development funding), worldwide efforts in PNCs have been growing exponentially. For example, the total number of hits for “polymer” and “nanocomposite” on SciFinder (Chemical Abstract Service (CAS) of the American Chemical Society) from 1988 to 2005 is >9400, where the yearly number has approximately doubled every 2 years since 1992 [4].

The possible applications of PNCs are huge and several PNC products have been already marketed: medical products (bandages, heart valves, etc.); food packaging material; electronic components; scratch-free paint; articles sporting goods; and wrinkle- and stain-resistant fabrics. Analysts have estimated that PNCs’ market is expected to reach USD 5.91 billion by 2018 [5]. Major revenues are forecast from large commercial opportunities, such as automobile, coatings, and packaging, where lower cost and higher performance resins would improve durability and design flexibility while lowering unit price. In this scenario, North America was the largest market for PNCs in 2011, accounting for approximately 40% of global demand. Europe accounted for a large portion of

PNC sales. Asia Pacific and the rest of the worlds (RoW) are expected to be strong future markets for nanocomposites with growing industrialization and infrastructure. Asia Pacific is expected to show the highest growth in demand and the market for nanocomposites is forecast to grow at a CAGR (compounded annual growth rate) of 18.8% between 2012 and 2018 [5].

Given the extensive variety of nanoparticles now commercially accessible, the potential combinations of polymers and nanoparticles, and thus the tailorability of the property suite, is essentially endless. The diversity in scientific investigation, technology advancement, processing innovations, and product development is staggering. A significant number of excellent review papers (e.g., clays, [6–16], CNTs [12, 13, 17–20], and spherical particles [21, 22]) and books [23–28] that chronicle and summarize the status of various nanoparticle–polymer combinations and the broad scientific and technological challenges still to be overcome are available. The most important issue for a complete exploitation of nanotechnology is to obtain a homogeneous dispersion of the particles in the matrix. This chapter is a critical review of the most significant method of PNC preparation (processability) and the most representative structure/morphology of polymer nanocomposites in relation to the dimensionality of the nanofiller.

20.1 OVERVIEW

PNCs are shown to have remarkable property enhancements relative to conventionally scaled composites. The transition from microparticles to nanoparticles leads to a number of

changes in physical properties. These changes are associated with the following three concepts:

1. As shown in Figure 20.1, the number of filler particles at a given volume fraction (3% in the figures) rapidly increases as the size of particles decreases. As a consequence, compared to conventional polymer composites with conventional fillers, which normally require loadings of 20% or more, the distances between nanoparticles are drastically reduced to the nanometer range, even when relatively low concentrations of nanofillers are used.
2. The sizes of the nanoparticles are comparable to the radius of gyration of the macromolecules in the polymer matrix, so that the morphological development of polymer matrix can be affected by the dispersed nanoparticles.
3. The nanoparticles provide ultrahigh specific surfaces (high surface area to volume ratio). As the surface area of a particle increases (Fig. 20.2), the interface becomes large and the amount of modified polymer interface relative to the total volume is significantly increased. This means that the portion of its constituent atoms at or near the surface increases exponentially, creating more sites for bonding, catalysis, or reaction with surrounding materials, resulting in improved properties such as increased strength or chemical and/or heat resistance.

Nanoparticles can be classified as a function of their geometry:

1. One dimension less than 100 nm (layered silicates: thickness about 1 nm);
2. Two dimensions less than 100 nm (nanotubes: diameter less than 100 nm);
3. Three dimensions less than 100 nm (spherical nanoparticles such as metal oxide, metal, etc.).

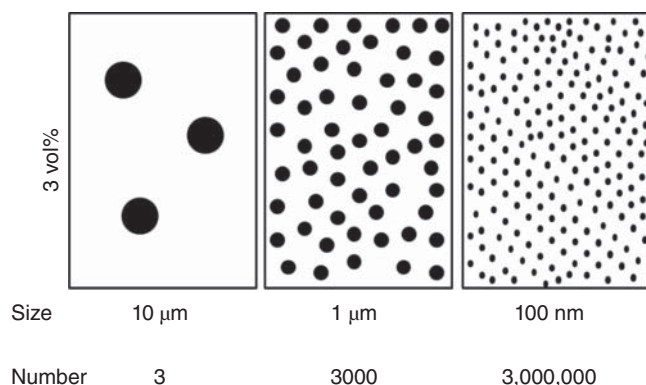


Figure 20.1 Illustration of how the number of particles (supposedly spherical) increases as the particle size decreases at a constant volume content.

The development of successful PNCs with a good balance of properties demands a high degree of coupling between processing and morphological and micromechanical control. That is, the balance of properties must be optimized. To this end, at least the essential morphological requirement should be met in the rational design of a nanocomposite: the uniform dispersion of nanofillers to avoid large agglomerates and high stress concentrations, and good or optimum interfacial bonding between them and the polymer matrix to achieve effective load transfer across the nanofiller and the matrix interface [29]. Unfortunately, it has been extensively reported that nanofillers are often dispersed in the polymer matrix in the form of agglomerates. This strong tendency to agglomerate significantly reduces their ability to bond with the polymer matrix because it reduces the contact area and decreases the effective aspect ratio (the aspect ratio of a nanoparticle is defined as the ratio between its sizes in different dimensions) of the reinforcement. Moreover, under external load the stress will be readily concentrated around such agglomerates, which, in turn, generally leads to the premature failure of the system and a worsening of the properties of the polymer. Good dispersion and distribution (Fig. 20.3) of nanoparticles in the polymer matrix are key factors for improved materials. The modification of the nanoparticles surface and/or the polymer, and the use of interfacial agents are the most used methodologies to achieve the best morphology [29].

20.2 NANOPARTICLES WITH ONE DIMENSION LESS THAN 100 NM (LAYERED SILICATES)

Layered silicates used in the synthesis of PNC are natural or synthetic minerals, consisting of very thin layers that are usually bound together with counterions. Their basic building blocks are tetrahedral sheets in which silicon is surrounded by four oxygen atoms, and octahedral sheets in which a metal like aluminum is surrounded by eight oxygen atoms. Therefore, in 1:1 layered structures (e.g., in kaolinite), a tetrahedral sheet is fused with an octahedral sheet, whereby the oxygen atoms are shared [30]. On the other hand, the crystal lattice of 2:1 layered silicates (or 2:1 phyllosilicates), consists of two-dimensional layers where a central octahedral sheet of alumina is fused to two external silica tetrahedra by the tip, so that the oxygen ions of the octahedral sheet also belong to the tetrahedral sheets, as shown in Figure 20.4.

The layer thickness is around 1 nm and the lateral dimensions may vary from 300 Å to several microns, and even larger, depending on the particulate silicate, the source of the clay, and the method of preparation. The aspect ratio of these layers (ratio length/thickness) is particularly high, with values greater than 1000 [7, 31–33].

Because, in their pristine state, layered silicates are only miscible with hydrophilic polymers, such as poly(ethylene oxide) (PEO) and poly(vinyl alcohol) (PVA), in order to render them miscible with other polymers, one must exchange the alkali counterions with a cationic-organic surfactant.

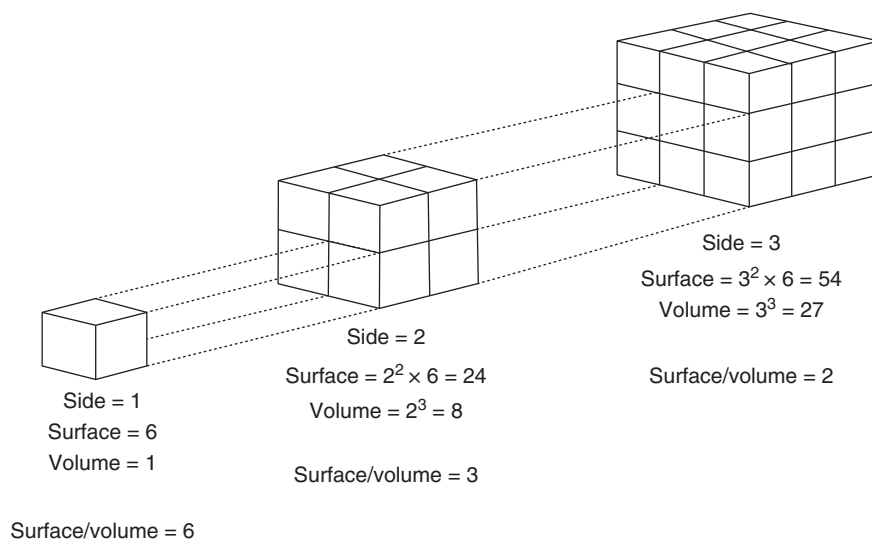


Figure 20.2 The dimensions, surface areas, and volumes of three different-sized cubes, which schematically represent particles, are compared here to show how surface area/volume ratio increases as the dimensions decrease.

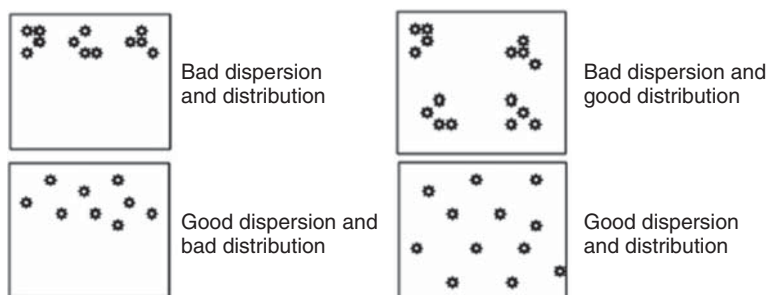


Figure 20.3 Schema of dispersion and distribution of nanoparticles in a polymer matrix, where dispersion and distribution are the level of nanoparticle conglomeration and homogeneity in the matrix, respectively.

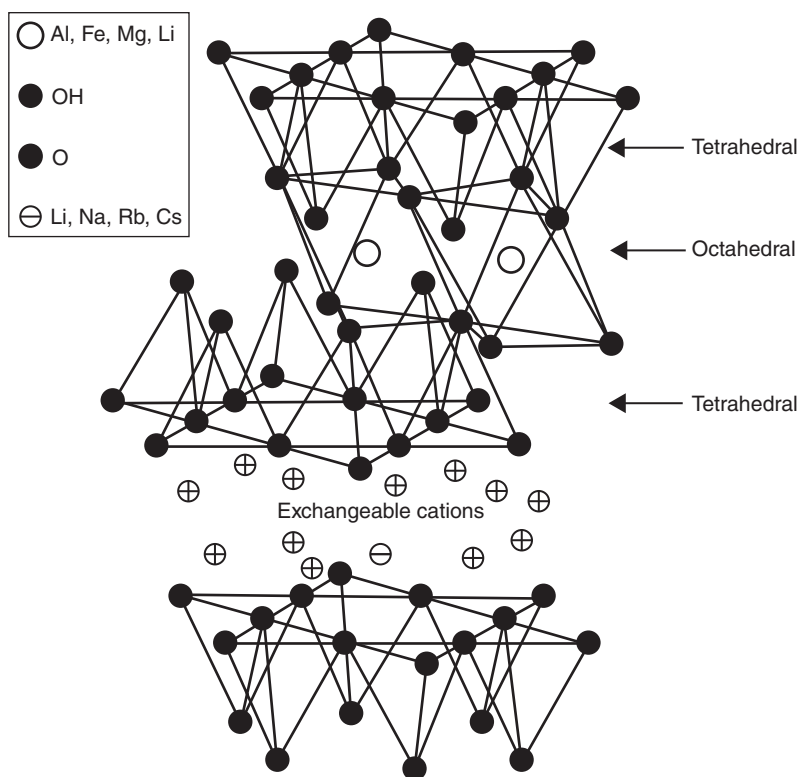


Figure 20.4 The structure of 2:1 layer silicates. Ray and Okamoto [8]. Reproduced with permission of Elsevier.

Alkylammonium ions are mostly used, although other salts can be used, such as sulfonium and phosphonium [7, 34, 35]. This can be readily achieved through ion-exchange reactions that render the clay organophilic [36]. In order to obtain the exchange of the salt ions with the cations in the galleries, water swelling of the silicate is needed. For this reason, alkali cations are preferred in the galleries because 2-valent and higher valent cations to prevent swelling by water. Indeed, the hydrate formation of monovalent intergallery cations is the driving force for water swelling.

Natural clays may contain divalent cations such as calcium and require exchange procedures with sodium prior to further treatment with ion salts [37]. The alkali cations, as they are not structural, can be easily replaced by other positively charged atoms or molecules, and thus are called exchangeable cations [38]. The organic cations lower the surface energy of the silicate surface and improve wetting with the polymer matrix [6, 39]. Moreover, the long organic chains of such surfactants, with positively charged ends, are tethered to the surface of the negatively charged silicate layers, resulting in an increase of the gallery height [37, 40], as shown in Figure 20.5. Conclusively, the surface modification both increases the basal spacing of clays and serves as a compatibilizer between the hydrophilic clay and the hydrophobic polymer [41].

Depending on the nature of components (polymer matrix, clay filler, and organic surfactant) and processing conditions, clay particles (organically modified or not) can be present in three configurations when incorporated in the polymer matrix (Fig. 20.6).

If the polymer is unable to intercalate into the galleries, a phase-separated composite is formed, whose properties are similar to that of traditional microcomposites; the poor interaction between the organic and the inorganic

component results in relatively poor mechanical performance. Unseparated layers, after introduction in a polymer matrix, are often referred to as tactoids [42]. On the other hand, an intercalated nanocomposite is obtained when extended polymer macromolecules diffuse between unchanged clay sheets, leading to a well-ordered multilayer structure of alternating polymeric and inorganic layers with a repeating distance of a few nanometers between them. The properties of this type of nanocomposites typically resemble those of ceramic materials. The most significant changes in physical properties are observed in exfoliated hybrids, where clay layers are separated and uniformly dispersed, thus maximizing the polymer–clay interactions [6, 43].

Both transmission electron microscopy (TEM) and wide-angle X-ray diffraction (WAXD) are essential tools [8] for evaluating PNC structures. With WAXD measurements, by monitoring the position, shape, and intensity of the basal reflections from the distributed layered silicates, the nanocomposite structure may be identified. In Figure 20.7 are reported TEM and WAXD for the different type of PNCs that can be generated. In the figure, the term flocculated nanocomposites indicates intercalated nanocomposites with hydroxylated edge–edge interaction between the silicate layers.

It is important to observe that the different structures can be generated simultaneously: an example [44] is reported in Figure 20.8.

20.3 NANOPARTICLES WITH TWO DIMENSIONS LESS THAN 100 NM (CARBON NANOTUBES)

CNTs were first reported by Iijima [45] in 1991, and the first PNCs using CNTs as a filler were reported in 1994 by Ajayan et al. [46] CNTs possess high flexibility [47], low mass density [48], and large aspect ratio (length/diameter of ~ 300 – 1000). CNT have a unique combination of mechanical, electrical, and thermal properties. Some nanotubes are stronger than steel, lighter than aluminum, and more conductive than copper. For example, theoretical and experimental results on individual single-walled carbon nanotubes (SWNTs) show extremely high tensile modulus [49] (640 GPa to 1 TPa) and tensile strength [50] (150–180 GPa). CNTs are long cylinders of covalently bonded carbon atoms. The ends of the cylinders may or may not be capped by hemifullerenes. There are two basic types of CNTs: single-walled carbon nanotubes (SWNTs) and multiwalled carbon nanotubes (MWNTs).

An SWNT can be considered as a single graphene sheet (graphene is a monolayer of sp^2 -bonded carbon atoms) rolled into a seamless cylinder along an (m,n) lattice vector in the graphene plane (Fig. 20.9). The carbon atoms in the cylinder have partial sp^3 character that increases as the radius of curvature of the cylinder decreases. The (m,n) indices determine the diameter and chirality, which are key parameters of a nanotube. Depending on the chirality (the chiral angle between hexagons and the tube axis), SWNTs can be either

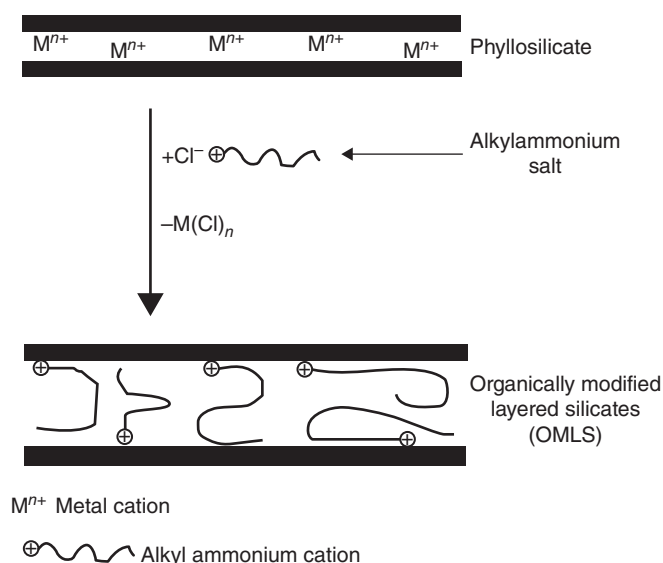


Figure 20.5 Schematic demonstration of clay organic modification. Zanetti et al. [37]. Reproduced with permission of John Wiley and Sons.

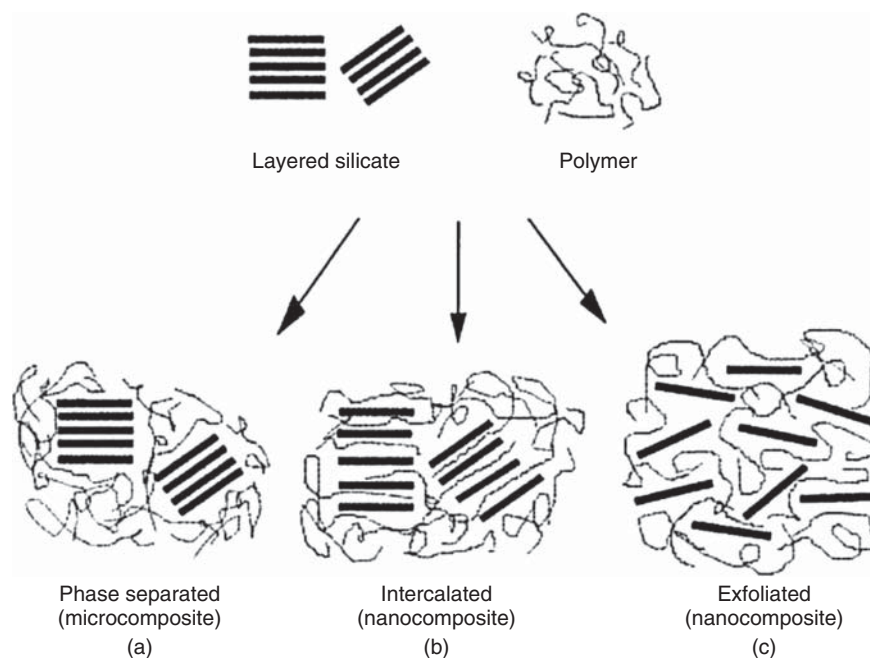


Figure 20.6 Possible polymer/layered silicate structures. Dubois [7]. Reproduced with permission of Elsevier.

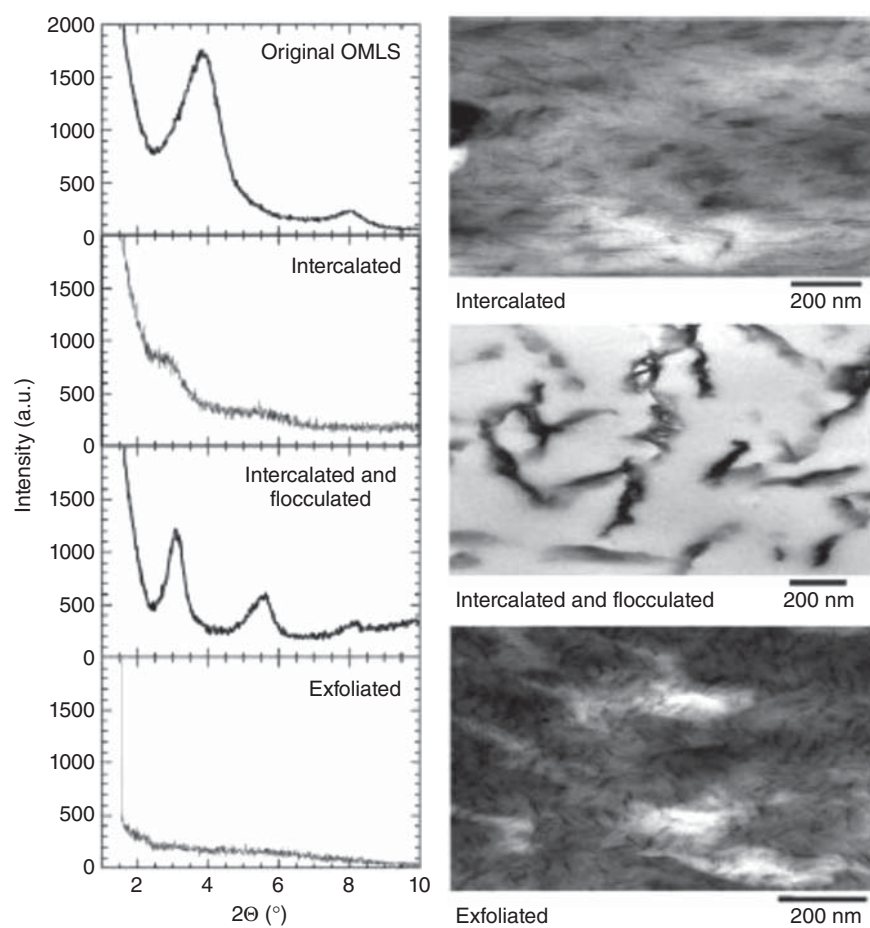


Figure 20.7 WAXD patterns and TEM images of three different PNC structures. Ray and Okamoto [8]. Reproduced with permission of Elsevier.

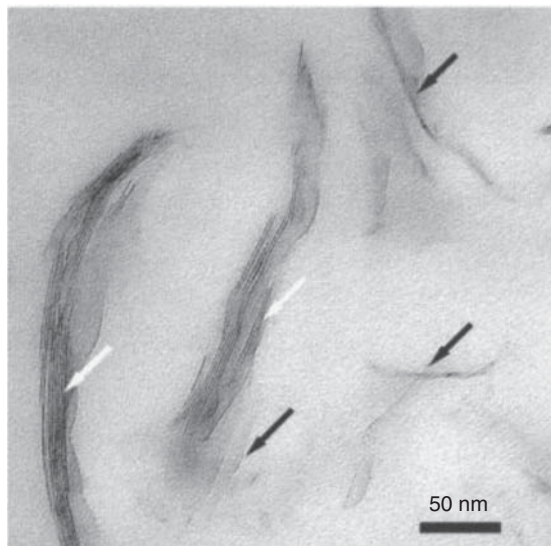


Figure 20.8 TEM image of an epoxy/clay nanocomposite that presents simultaneously an intercalated (white arrows) and exfoliated (black arrows) structure. Miyagawa et al. [44]. Reproduced with permission of John Wiley and Sons.

metals or semiconductors, with band gaps that are relatively large (~ 0.5 eV for typical diameter of 1.5 nm) or small (~ 10 meV), even if they have nearly identical diameters [51]. For same-chirality semiconducting nanotubes, the band gap is inversely proportional to the diameter. Thus, there are infinite possibilities in the type of carbon tube “molecules,” and each nanotube could exhibit distinct physical properties.

MWNTs consist of nested graphene cylinders coaxially arranged around a central hollow core with interlayer separations of ~ 0.34 nm, indicative of the interplane spacing of graphite [52]. A special case of MWNT is double-walled nanotubes (DWNTs) that consist of two concentric graphene cylinders. DWNTs are expected to exhibit higher flexural modulus than SWNTs, due to the two walls and higher toughness than regular MWNTs because of their smaller size [53].

Presently, MWNTs and SWNTs are mainly produced by three techniques, each having their own nuances: arc discharge, laser ablation, and chemical vapor decomposition (CVD). A number of reviews [52, 54] are available on these production techniques. Arc discharge and laser ablation methods involve the condensation of hot gaseous carbon atoms generated from the evaporation of solid carbon. In CVD, a gaseous carbon source (hydrocarbon, CO) is decomposed catalytically, and the nanotubes are deposited on a substrate or they grow from a substrate. Compared with arc and laser methods, CVD might offer more control over the length and structure of the produced nanotubes. As mentioned earlier, at present, all known preparations of CNTs give mixtures of nanotube chiralities, diameters, and lengths along with different amount and type of impurities. These parameters vary significantly both within a sample and between samples from different batches and laboratories. Thus, it is very

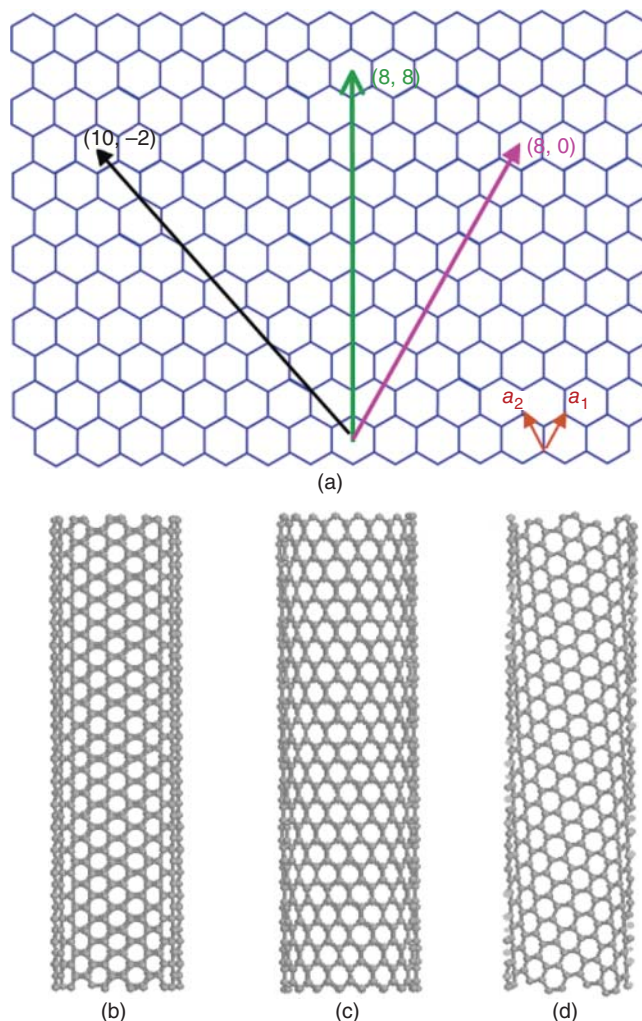


Figure 20.9 (a) Schematic honeycomb structure of a graphene sheet. Single-walled carbon nanotubes can be formed by folding the sheet along lattice vectors. The two basis vectors a_1 and a_2 are shown. Folding of the (8,8), (8,0), and (10,−2) vectors lead to (b) armchair, (c) zigzag, and (d) chiral tubes. Dai [51]. Reproduced with permission of American Chemical Society.

difficult to conduct reproducible control experiments with these inconsistent nanofillers and virtually impossible to compare results between different researchers.

Another great challenge in nanotube/polymer composites is the efficient translation of nanotube properties, both into the polymer matrix and between nanotubes.

This is why CNTs are functionalized with organic molecules before/during their mixing with a polymer matrix. Functionalization of CNTs can serve to not only improve their solubility and, therefore, dispersion but also to create attractive van der Waals interactions between the polymer matrix and the nanotube addend [55]. Several research groups have reported successful functionalization reactions for SWNT and MWNT CNTs [55–75].

These reactions may roughly be divided into two categories: a direct attachment of functional groups to the graphitic

surface and the use of the nanotube-bound carboxylic acids. The two categories of functionalization reactions may have different effects on the structures and intrinsic properties of CNTs.

In the first category, Margrave, Smalley, and coworkers reported the fluorination of SWNTs [57,58]. In various alcohol solvents, these functionalized SWNTs were solvated as individual tubes, making it possible to carry out further solution chemistry. Pekker and coworkers reported the hydrogenation of CNTs via the Birch reduction in ammonia [70]. Other examples include the derivatization of small-diameter (ca. 0.7 nm) SWNTs reported by Tour and coworkers [68, 69]; the interactions with anilines reported by Wilson and coworkers [67]; the reactions with nitrenes, carbenes, and radicals reported by Hirsch and coworkers [74]; and the 1,3-dipolar addition reported by Prato and others [75].

In the second category of functionalization reactions, the nanotube-bound carboxylic acids come from intrinsic or induced defects. The latter refer to the creation of terminal carbons in the shortening of nanotubes, which upon oxidation are converted to carboxylic acids [56, 76, 77]. Haddon and coworkers first reported the use of the acid groups for attaching long alkyl chains to SWNTs via amide linkages [56] or carboxylate-ammonium salt ionic interactions [59]. Sun and coworkers showed that the esterification of the carboxylic acids can also be applied to functionalize and solubilize nanotubes of any length [60, 65, 66]. An advantage with the ester linkages is that they can be facilely defunctionalized via acid- or base-catalyzed hydrolysis, allowing the recovery of CNTs from the soluble samples [66]. The nanotube-bound carboxylic acids are the sites to attach a variety of functional groups for the solubilization of both shortened and full-length CNTs.

20.4 NANOPARTICLES WITH THREE DIMENSIONS LESS THAN 100 NM (METAL, METAL OXIDE)

Isodimensional nanoparticles used for preparation of PNCs are mainly oxide and metal oxide. The most common are silver (Ag), gold (Au), zinc oxide (ZnO), silica (SiO₂), titanium dioxide (TiO₂), and alumina (Al₂O₃). A comprehensive coverage of all the different kinds of isodimensional nanoparticles would be beyond the scope of this review. Instead, this paragraph focuses, principally, on the most commonly used isodimensional nanoparticles, that is, silver, zinc oxide, titanium dioxide, and silica.

Ag nanoparticles are clusters of Ag atoms that range in diameter from 1 to 100 nm and are attracting interest because their most innovative characteristics are associated with their increased antimicrobial properties [22, 23, 78, 276–278]. Owing to these antimicrobial properties, Ag nanoparticles are currently the most commonly used nanoparticles in consumer and industrial applications. The use of Ag as an antimicrobial agent, however, is not a new concept. Ag pots and coins were

used in ancient times to keep water sterile and, in particular, Ag-nitrate (AgNO₃) has been used since the seventeenth century as an essential multipurpose medicinal product [79, 80]. Examples of recent consumer applications using Ag as an antimicrobial agent consist of food supplements, materials for food packaging, coatings on medical devices, water disinfectants, air filters, electronic appliances, odor-resistant textile fabrics, and cosmetic products such as deodorants [81].

In general, there are two strategies to produce metal nanoparticles. One is the so-called bottom-up method, where nanostructures are obtained through assembly of smaller, basic units into larger structures. The other is the top-down method, which starts from bulk material, whose size is decreased during the process. Common top-down techniques include photolithography and electron beam lithography [82, 83].

The most extensively studied method for synthesizing Ag nanoparticles is the controlled reduction of Ag salts [84, 85]. This reduction is often accomplished using a chemical reducing agent [86–88] but also via electrochemical [89] or photochemical reduction [90, 91]. This wet-chemical method, which involves Ag salts reduced (in one or more steps) to metal Ag, can be steered by many different parameters, such as the choice of the reducing agent, the concentrations of reagents, temperature, pH, mixing rate, and reaction time [90]. In order to tailor the particle properties, many studies have focused on the generation of different shapes, such as rods, wires, cubes, thin plates, or bipyramids [92, 93]. In addition, substantial effort has been devoted to the construction of core-shell structures, such as Ag@Au (silver core particles with a gold shell) or Ag@SiO₂ nanoparticles [94, 95]. In general, strong reducing agents result in the formation of smaller NPs, whereas weaker reducing agents usually lead to larger Ag particles [90]. Another method makes use of a Tollens process to produce Ag nanoparticles in a one-step method [90]. This reaction has been long used in the electrodeless deposition of Ag to generate reflective mirrors on solid supports and combines reducing sugars and Ag ammoniacal solutions [86]. The obtained particle size distributions are generally narrow. The Tollens method is an example of a green synthesis of Ag nanoparticles [92].

Among metal oxides, nano-ZnO, nano-TiO₂, and nano-SiO₂ have been objects of an intense research. Nano-ZnO and nano-TiO₂ display antibacterial activity and act as UV blocking agents. Certain forms of TiO₂ are also photocatalytic, and have been incorporated in products such as glazing and cement to impart self-cleaning properties. Interest also extends to the electronic properties of TiO₂ and it is a key component of dye-sensitized solar cells. Interest in nanoscale ZnO extends to its electronic properties, which are being exploited for use in such products as transparent electrodes, LEDs, lasers, and piezoelectric materials. Nano-SiO₂ possess ordered porous structures with extremely high surface areas (>1000 m²/g), well-defined and tunable pore sizes (1.5–10 nm), and can be easily functionalized by many approaches. Recently, several groups have taken them as suitable reservoirs of drugs and biomolecules for biomedical applications [96].

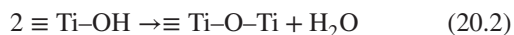
Many different methods have been reported for the production of metal oxide nanoparticles. The most common processes have been developed to synthesize oxide are chemical solution decomposition (CSD) [97], spray pyrolysis [98], CVD [99–101], two-step wet chemical method [102], sol–gel [103], ultrasonic irradiation [104, 105], and ethanol thermal and hydrothermal method [106]. Literature revealed that sol–gel is the most commonly used method for the preparation of metal oxide nanoparticles. It facilitates the synthesis of nanometer-sized crystallized metal oxide powder of high purity at a relatively low temperature [103].

The sol–gel chemistry is based on the hydrolysis and condensation reactions of metal alkoxides, $M(OR)_4$: where M is a metal ion and R an alkyl group (R = Me, Et). In the case of titanium, hydrolysis leads to the formation of Ti–OH groups:

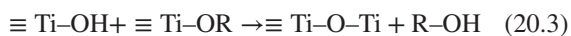


The condensation reactions lead to the formation of oxo bridges, Ti–O–Ti represented by the following reactions:

Oxalation or water elimination



And alcoxylation or alcohol elimination



Finally, the overall generalized reaction can be written as



Many factors influence the kinetics of the hydrolysis and condensation reactions in the sol–gel process, which include the water/alkoxide ratio, temperature, the nature of solvent, and so forth [107].

The functionalization of these particles is always required for preparing a PNC; in fact nanoparticles have a strong tendency to agglomerate and, consequently, the so-called nanoparticle-filled polymers sometimes contain a number of loosened clusters of particles (Fig. 20.10a) and the final material exhibits properties even worse than conventional particle/polymer systems.

Noble metal nanoparticles were mainly modified by thiols [109–115], disulfides [116], amines [111–113, 117, 118], nitriles, carboxylic acids, and phosphines [113, 118, 119]. The use of organosulfur compounds for modifying noble metal nanoparticles is one of the more developed methods, because organosulfur groups strongly coordinate to various metals, such as Ag, Cu [120], Pt, Hg, Fe, or Au. Sulfur possesses a huge affinity for metal surfaces, and organosulfur compounds thus will adsorb spontaneously [121].

Various organic compounds are potential metal oxide modifiers, among them, as for metal nanoparticles, thiols, carboxylic acids, and amines [122–124]. However, amines and thiols are relatively rarely used; an example is thiourea to modify SnO_2 surfaces [122]. The main compounds used for modifying metal oxide nanoparticles are phosphonates or silanes. Among silanes, the most used include alkoxysilane, $\equiv Si-OR$ (R = alkyl), hydrogenosilane, $\equiv Si-H$, or chlorosilane, $\equiv Si-Cl$, reagents. As a matter of fact, there is a broad variety of commercially available silanes, and others are readily synthesized or modified. One of the main advantages of silanes is that they can bear numerous functionalities, for example, amino, cyano, carboxylic acid, epoxy groups, and so on [123, 125–132]. The postsynthesis grafting of silyl groups on a metal oxide surface is quite easy. Moreover, silanes can also be used for the one-step method, where the silane is introduced at the same time as the metal oxide precursors [110]. This method is applicable to numerous metal oxides such as SiO_2 , Al_2O_3 , TiO_2 , SnO_2 , ZrO_2 , V_2O_5 , and so on.

An irradiation grafting method was applied for the modification of nanosilica in the preparation of polypropylene

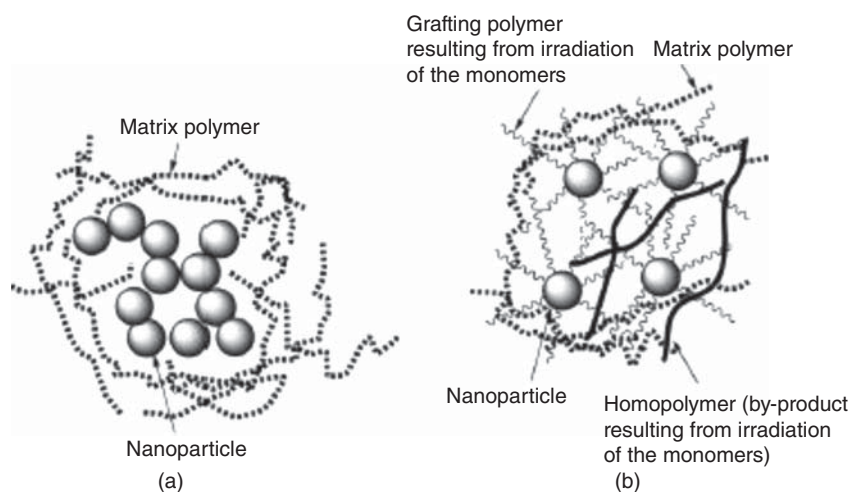


Figure 20.10 Schematic drawings of (a) agglomerated nanoparticles dispersed in a polymer matrix and (b) the possible structure of grafted nanoparticles dispersed in a polymer matrix. Rong et al. [108]. Reproduced with permission of Elsevier.

(PP)/nanosilica composites have been reported by Zhang and coworkers [108,133]. Through irradiation grafting polymerization, nanoparticle agglomerates turned into a nanocomposite microstructure (comprising the nanoparticles and the grafted, homopolymerized secondary polymer; Fig. 20.10b), which in turn built up a strong interfacial interaction with the surrounding, primary polymeric matrix during the subsequent mixing procedure.

20.5 PREPARATIVE METHODS

The properties of a polymer nanocomposites and their improvement with respect to the traditional composites depend on the structure and morphology developed during the preparation step. In turn, structure and morphology depend on preparation conditions.

At present, the most used methods to prepare polymer nanocomposites are (i) solution processing; (ii) *in situ* polymerization; (iii) melt processing; and (iv) *in situ* sol-gel technology.

20.5.1 Solution Processing

In this approach, a dispersion of nanoparticles in a suitable solvent and polymer are mixed in solution. The PNC is formed by precipitation or by evaporation of the solvent, usually under vacuum.

When layered silicates are used, the weak forces that stack the layers together can be easily dispersed in an adequate solvent. After the clay/organoclay has swollen in the solvent, the polymer is added to the solution and intercalates between the clay layers (in this case, the method is called *intercalation of polymer from solution*).

Even though this technique has been mostly used with water-soluble polymers, such as PEO, polyvinyl ether (PVE), polyvinylpyrrolidone (PVP), and poly(acrylic acid) (PAA) [134–141], intercalation from nonaqueous solutions has also been reported [142–145]. For example, high-density polyethylene (HDPE)-based nanocomposites have been prepared by dissolving HDPE in a mixture of xylene and benzonitrile with dispersed organomodified layered silicates (OMLSs). The nanocomposite was then recovered by precipitation from tetrahydrofuran (THF) [143]. Polystyrene (PS)/OMLS-exfoliated nanocomposites have also been prepared by the solution intercalation technique, by mixing pure PS and organophilic clay with adsorbed cetyl pyridium chloride [146]. Similarly, several studies have focused on the preparation of polylactide (PLA)-layered silicate nanocomposites using intercalation from solution.

The first attempts by Ogata et al. [142] involved dissolving the polymer in hot chloroform in the presence of OMLS. However, TEM and WAXD analyses revealed that only microcomposites were formed and that an intercalated morphology was not achieved. In a later study, Krikorian and Pochan [147] prepared PLA nanocomposites using dichloromethane as the

polymer solvent and as the OMLS dispersion medium. The authors obtained intercalated or exfoliated nanocomposites, depending on the type of OMLS used. That is, exfoliated nanocomposites were formed when diols were present in the organic modifier of the clay, due to the favorable enthalpic interaction between these diols and the C=O bonds in the PLA backbone. Chang et al. [148] reported the preparation of PLA-based nanocomposites with different kinds of OMLS via solution intercalation using *N,N*-dimethylacetamide (DMA).

In the case of polymeric materials that are infusible and insoluble even in organic solvents, the only possible route to produce nanocomposites with this method is to use polymeric precursors that can be intercalated in the layered silicate and then thermally or chemically converted to the desired polymer [149]. It is important to note that, in using this method, intercalation only occurs for certain polymer/clay/solvent systems, meaning that for a given polymer one has to find the right clay, organic modifier, and solvents [150].

When other nanoparticles (CNTs, metal oxide, etc.) are used, their dispersion in a solvent by simple stirring is very difficult. A high-power ultrasonication process is more effective in forming a dispersion of particles. Ultrasonic wave and mechanically stirring play important roles in the formation of the composites with a uniform particle size. The chemical effects of ultrasound are associated with the rapid (microsecond timescale), violent collapse of cavitation bubbles created as the ultrasonic waves pass through a liquid medium [151]. Sonochemical theory and the corresponding studies suggested that ultrasonic cavitation can generate a high local temperature of 5000 K and a local pressure of 500 atm [152], which is a very rigorous environment.

For example, Li et al. [153] used a simple solution-precipitation technique to improve the dispersion of CNTs in a polycarbonate solution by sonication at a frequency of 20 kHz for 10 min. They showed that the CNTs were uniformly dispersed in polycarbonate matrix on its consolidation. Safadi et al. [154] dispersed MWNTs in PS using ultrasonication and dismembrator at 300 W for 30 min. Uniform dispersions of CNTs in PS were achieved by using sonication. Recently, Cho and coworkers successfully prepared polyurethane (PU)/MWNT composites with better dispersion of CNTs up to 20 wt% in PU [155].

Solution processing has been used for preparation of many different polymer/CNTs such as PU/CNT [156], PS/CNT [157–159], epoxy/CNT [158, 160, 161], poly(vinyl alcohol) (PVA)/CNT [162], polyacrylonitrile (PAN)/CNT [163], and polyethylene (PE)/CNT [164].

By this method also, polymer/silica nanocomposites have been fabricated [165, 166]. Merkel et al. discovered that the addition of nanometer-sized fumed silica particles to certain high-free volume, glassy polymers could systematically increase gas permeability. Such high-permeability polymers included poly(4-methyl-2-pentyne) (PMP), poly[1-(trimethylsilyl)-1-propyne] (PTMSP), and poly(2,2-bis(trifluoromethyl)-4,5-difluoro-1,3-dioxole-*co*-tetrafluoroethylene) [107].

van Zyl et al. [167] reported the preparation of PA/silica nanocomposites via solution processing. Nylon 6 was first dissolved in formic acid, the pH was controlled at ca. 2, and the silica sol with particle sizes of 10–30 nm was added to the nylon solution and stirred gently at room temperature. The solution was then casted, and the solvent was evaporated. The nanocomposite was examined with TEM, which revealed that the silica particles were well dispersed and nonaggregated.

20.5.2 In situ Polymerization

In this method, the nanoparticles are dispersed in monomer followed by polymerization. This method is useful for the preparation of composites with polymers that cannot be processed by solution or melt mixing, for example, insoluble and thermally unstable polymers [168, 169].

In situ polymerization was the first method used to synthesize polymer–clay nanocomposites based on polyamide (PA) 6. In this technique, the modified layered silicate is swollen by a liquid monomer or a monomer solution. The monomer migrates into the galleries of the layered silicate, so that the polymerization reaction can occur between the intercalated sheets. The reaction can be initiated either by heat or radiation, by the diffusion of a suitable initiator or by an organic initiator or catalyst fixed through cationic exchange inside the interlayer before the swelling step by

the monomer. Polymerization produces long-chain polymers within the clay galleries. Under conditions in which intra- and extra-gallery polymerization rates are properly balanced, the clay layers are delaminated and the resulting material possesses a disordered structure [33, 170]. The method has been applied for the preparation of nanocomposites based on thermoplastic polymers other than polyamides, including polymethylmethacrylate (PMMA) [171, 172], (PS) [173], polybenzoxale (PBO) [174], polyolefins (PP and PE), [175–178], and polyethylene terephthalate (PET) [179].

In situ polymerization has been widely used for the preparation of PMMA-CNT composites [180–183]. Conducting polymers are attached to CNT surfaces by *in situ* polymerization to improve the processability and electrical, magnetic, and optical properties of CNTs [184–187]. PU/MWNT [188, 189] and PA/MWNT [190] composites were also synthesized by this method.

An interesting example is proposed by Sahoo et al. [191], where polypyrrole (PPy)-coated MWNTs were synthesized by *in situ* polymerization of pyrrole on MWNT-COOH. The possible interaction of hydrogen bonding between PPy chain and MWNT-COOH is reported in Figure 20.11(1). They investigated the effect of the monomer concentration on the coating and properties of the resulting complex nanotubes. By changing the pyrrole/MWNT ratio, the layer thickness of PPy could be easily controlled in MWNT-PPy complex nanotubes, as shown in Figure 20.11(2).

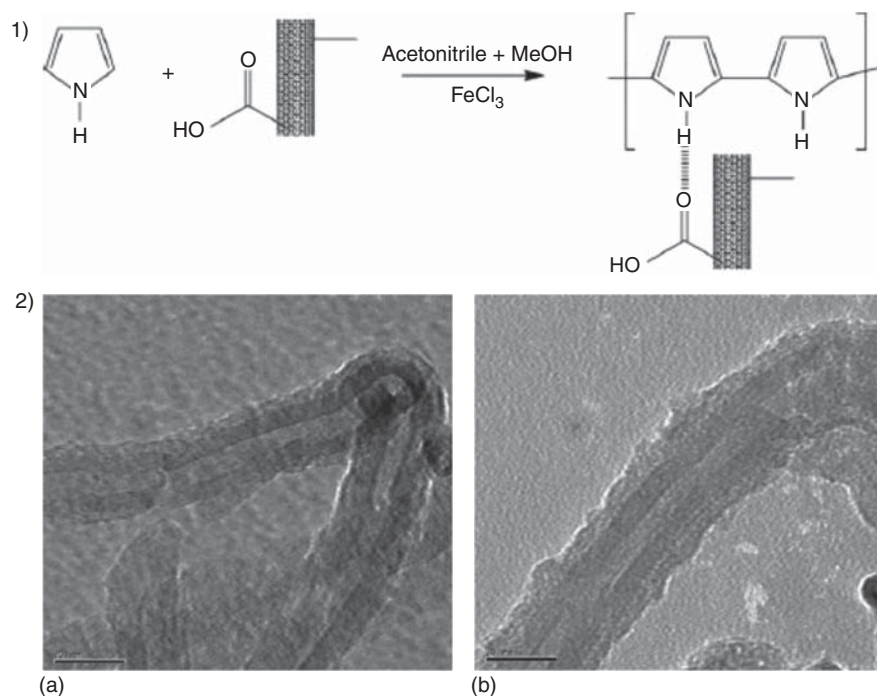


Figure 20.11 (1) Possible interaction of hydrogen bonding existed between for PPy chain and MWNT-COOH. (2) TEM micrographs of PPy-coated MWNTs: by changing the monomer concentration ((a) MWNT:PPy = 1:2; (b) MWNT:PPy = 1:5), the layer thickness of polymerized PPy changed, resulting in different electrical properties. Sahoo et al. [191]. Reproduced with permission of Elsevier.

20.5.3 Melt Processing

Melt processing is a common and simple method, particularly useful for thermoplastic polymers. This technique consists of blending the nanoparticles with the polymer matrix in the molten state. The nanoparticles are mechanically dispersed into a polymer matrix using a high-temperature and high-shear-force mixer or compounder [192].

When technologically important polymer/layer silicate nanocomposites are to be prepared, both *in situ* polymerization and intercalation from solution are limited because neither a suitable monomer nor a compatible polymer–silicate solvent system is always available. Moreover, they are not always compatible with current polymer processing techniques. These disadvantages drive the researchers to the direct melt intercalation method, which is the most versatile and environmentally benign among all the methods of preparing polymer–clay nanocomposites [6]. The advantages of forming nanocomposites by melt processing are quite appealing, rendering this technique a promising new approach that would greatly expand the commercial opportunities for nanocomposite technology [193–196]. If technically possible, melt compounding would be significantly more economical and simpler than *in situ* polymerization. It minimizes capital costs because of its compatibility with existing processes. That is, melt processing allows nanocomposites to be formulated directly using ordinary compounding devices such as extruders or mixers, without the necessary involvement of resin production. Therefore, it shifts nanocomposite production downstream, giving end-use manufacturers many degrees of freedom with regard to final product specifications (e.g., selection of polymer grade, choice of organoclay, level of reinforcement, etc.). At the same time, melt processing is environmentally sound because no solvents are required [193, 195]; and it enhances the specificity for the intercalation of polymer, by eliminating the competing host–solvent and polymer–solvent interactions [2]. Thus, the majority of thermoplastic polymers, including PA [32, 195, 197, 198], PET [199] (and recycled PET [200]), EVA [201, 202], thermoplastic PU [203], polyolefins [204, 205], PLA [206–208], polycaprolactone (PCL) [209, 210], and so on, have been used to prepare polymer/clay nanocomposites by this method.

Melt processing has been successfully applied for the preparation of different polymer–CNT composites such as PP/CNT [211–213], PE/CNT [212–216], polycarbonate (PC)/CNT [217–219], PMMA/CNT [220–223], polyoxymethylene/CNT [224], polyimide/CNT [225], PA6/CNT [226, 227], and so on. The shear forces from the mixers/extruders should help break nanoparticle aggregates or prevent their formation. Unfortunately, the dispersion of CNTs in a polymer matrix is quite poor compared to the dispersion that may be achieved through solution mixing. In addition, the CNTs must be lower due to the high viscosities of the composites at higher loading of CNTs.

Melt processing has been used also for polymer and polymer blend/silica nanocomposites. In particular, extensive studies are reported for PP [228], PP-based copolymer [229], PE [230], PE-based copolymer [231, 232], PS [233–235], PMMA [234, 235], PC [234], PC-based copolymer [236], polyethylene naphthalate (PEN) [237], perfluoropolymer [238], PET [239–241], PA [242], polyvinyl acetate (PVAc) [243], copolyetherester [244], styrene-butadiene rubber [245, 246], ethylene vinyl acetate (EVA) [247], PET/PS [248], PLLA [249], and many others.

20.5.4 In situ Sol–Gel Technology

An extensive literature survey reveals that almost all polymer–ceramic nanocomposites were prepared through mechanical blending of nanosized particles, polymer, and salt in a compatible solvent. Aggregation of these particles stemming from their high surface energy is a perennial problem that undermines the efficacy of the ceramic fillers. A simple and effective method to overcome such a problem is the sol–gel process, wherein the nanosized ceramic fillers are precipitated *in situ* in the polymer matrix through a series of hydrolysis and condensation reactions of suitable precursors. Thus, the ceramic fillers are uniformly distributed in the polymer and exhibit excellent properties.

These organic–inorganic hybrids are generally classified into two broad families: in type I nanocomposites, the interactions between the hosting matrix and the entrapped species are weak and based on hydrogen bonds and van der Waals forces; whereas in type II nanocomposites, the organic and inorganic components are connected through stronger chemical bonds.

The successful implementation of this versatile process in the synthesis of siloxane-derived composite polymer (i.e., polymer/silica nanocomposites) electrolytes has appeared in several recent reports [250–252]. The sol–gel method has already been reported for the preparation of nanocomposite-based silicone rubber [253, 254], polyisobutylene [255], and several diene rubbers [256–261]. Nevertheless, this method is suitable usually for amorphous polymers (rubbers and elastomers) as it is based on the polymer swelling by the inorganic precursor solution. Regarding polyolefin matrix, Sun et al. [262] reported the preparation of polypropylene/silica nanocomposites via a two-step method: (i) diffusion of the tetraethoxysilane (TEOS) through the polypropylene (PP) matrix using supercritical carbon dioxide as a carrier and swelling agent, (ii) hydrolysis/condensation reactions of the precursor molecules confined in polymer network. Jain et al. [263, 264] developed a new process, combining the sol–gel method with the solid-state modification consisting of grafting vinyl triethoxysilane (VTEOS). Following this, they prepared PP/silica nanocomposites, varying the degree of covalent interaction between silica nanofiller and matrix. Dou et al. [265] reported a preparation of PP/silica

nanocomposite by *in situ* sol–gel process using hyperbranched polyethoxysiloxane.

Several works have been published on the efficiency of the sol–gel process in the production of high-quality titanium dioxide from titanium alkoxide precursor [266–270].

20.6 STRUCTURE AND MORPHOLOGY OF POLYMER NANOCOMPOSITES

Good dispersion and distribution (and in the case of clay nanoparticles an exfoliated structure) are the key factors for improving properties in a nanocomposites.

The different structure and morphologies that can be generated during the preparation of a nanocomposite depend on many parameters. The most important are method of preparations, nature of the particle functionalization, particle concentrations. A comprehensive coverage of all the different effect obtained by changing the parameters would be not possible and beyond the scope of this review. Anyway some examples in this paragraph are reported.

In Figure 20.12, different dispersions of CNTs in epoxy matrix, fabricated using different techniques are shown by TEM.

The techniques used for preparation of nanocomposites (sonication in water bath, shear mixing, probe sonicator, probe sonicator,

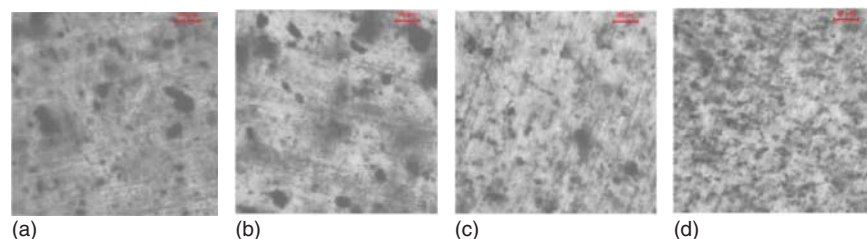


Figure 20.12 Dispersion of CNTs in nanocomposites fabricated using different dispersion techniques (a: sonication in water bath; b: shear mixing; c: probe sonicator; d: calendaring; scale bar = 20 μm). Peng-Cheng et al. [271]. Reproduced with permission of Elsevier.

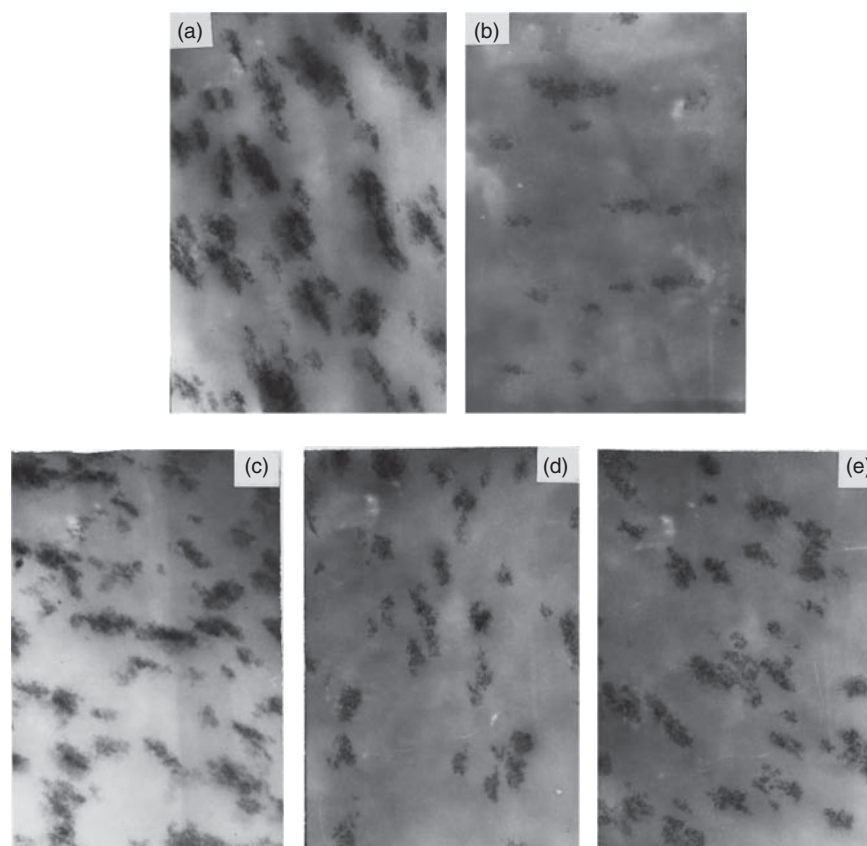


Figure 20.13 TEM micrographs of PP (MI = 6.7 g/10 min)-based nanocomposites filled with (a) SiO₂ as-received (content of SiO₂ 1.96 vol%), (b) SiO₂-g-PS (content of SiO₂ 1.96 vol%), (c) SiO₂-g-PS (content of SiO₂ 6.38 vol%), (d) SiO₂-g-PMMA (content of SiO₂ 1.96 vol%), and (e) SiO₂-g-PMMA (content of SiO₂ 6.38 vol%). Magnification 2×10^4 . Rong et al. [108]. Reproduced with permission of Elsevier.

and calendaring) influences the dispersion of CNTs and, to a large extent, the mechanical properties of CNT/polymer nanocomposites. The best CNT dispersion was achieved by employing calendaring machine (Fig. 20.12d), and the nanocomposites fabricated using this technique exhibited the best flexural properties among the four techniques used [271,272].

In the work of Rong et al. [108], various polymers were grafted on the surface of nanoscale silica filler particles through the simultaneous irradiation polymerization technique. In this way, the modified nanoparticles can be more effectively utilized in thermoplastics (such as PP) than conventional particulate fillers, when using the same direct compounding technology.

Morphology of PP/SiO₂ is reported in Figure 20.13, where different polymers (PS and PMMA) are grafted on the silica surface by irradiation treatment. Two different concentrations of SiO₂ are considered (1.96 and 6.38 vol%). From the figure it is clear that higher filler loading is detrimental to silica uniform dispersion in the polymer matrix. In fact, the size of the dispersed phases increases with increasing filler content in the nanocomposites. It should be noted that the dispersed phases in

the nanocomposites illustrated in Figure 20.13b–e are clearly smaller than the untreated SiO₂.

As a result of these phenomena, reinforcing and toughening effects of the nanoparticles on polymeric materials could be fully brought into play. In fact, the tensile properties, such as strength, modulus, and elongation at break of PP were improved simultaneously when the modified nanoparticles were incorporated.

In the work of Bahloul [270], *in situ* generation of TiO₂ inorganic particles in a semicrystalline polymer (PP) is reported. Titanium n-butoxide was mixed with PP under molten conditions in a corotating twin screw extruder. At the die exit of the extruder, the extent of the hydrolysis was higher than 70% and fine particles dispersed in PP were observed by TEM analysis. Figure 20.14a and b shows a fine dispersion of few nanometers of TiO₂ particles, whereas Figure 20.14c, which presents the morphology of nanocomposites with higher percentage of silica particles, indicates larger scale structure (aggregates) at the nanometric scale.

When melt-mixing process is considered for preparation of nanocomposites, the effects of operating and processing conditions on the morphology of nanocomposites, and therefore

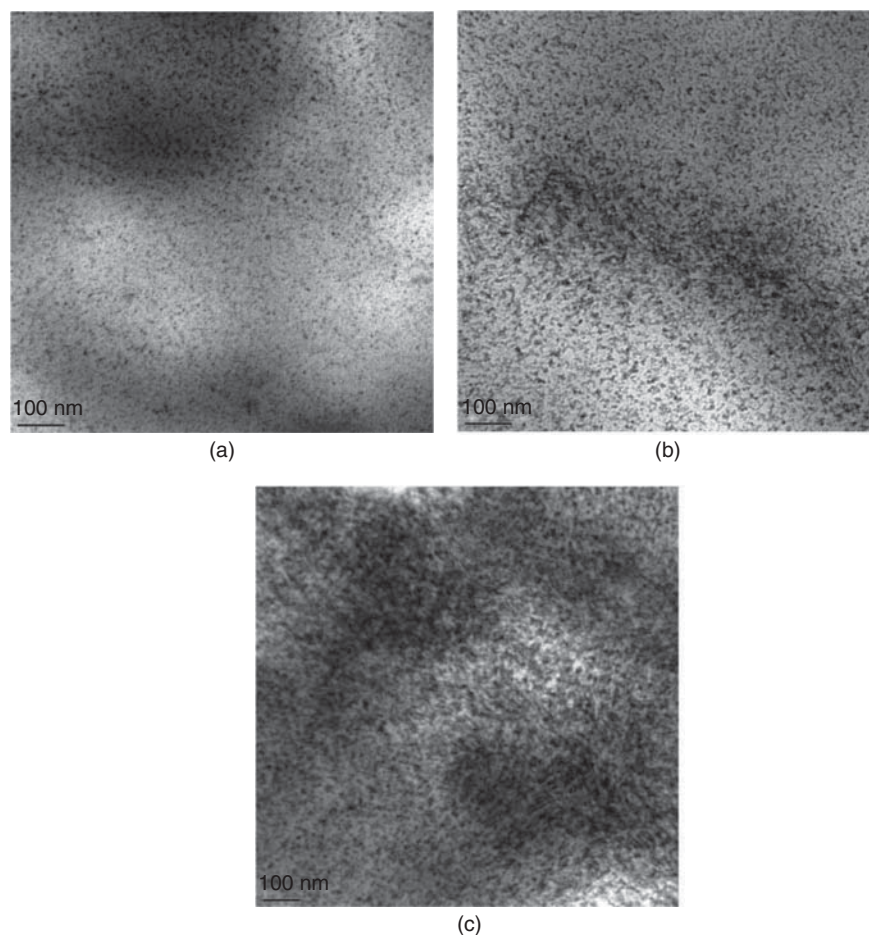


Figure 20.14 TEM of nanocomposites containing different amounts of nano-TiO₂: (a) PP-2.9%TiO₂, (b) PP-6%TiO₂, and (c) PP-9.3%TiO₂. Bahloul et al. [270]. Reproduced with permission of John Wiley and Sons.

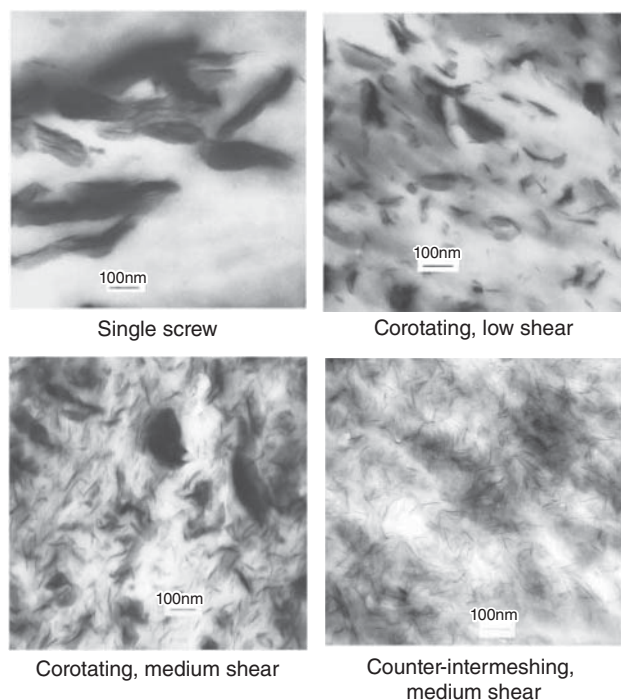


Figure 20.15 Transmission electron micrographs for selected nanocomposites prepared from clay and PA6 using different extruders and screw configurations. Dennis et al. [273]. Reproduced with permission of Elsevier.

on their properties, has to be considered. Dennis et al. [273] have studied the effect of the processing conditions on the formation of nanocomposites from PA6 and organoclay by using several combinations of extruders and screw configurations. In particular, they have used a single-screw extruder and two twin-screw extruders: corotating twin-screw extruder and counterrotating twin-screw extruder. For each screw extruder type, the shear intensity of mixing was changed by changing the screw configurations. Excessive shear intensity or back-mixing apparently causes poorer delamination and dispersion and the best results are obtained by using a counterrotating twin screw extruder (Fig. 20.15).

In another work, Lertwimolnun and Vergnes [274] have prepared PP/organoclay nanocomposites by using a twin-screw extruder and have found that the state of intercalation, interpreted by interlayer spacing, is globally unaffected by processing parameters. A complete intercalated structure is obtained, even at high feed rate, in a very short period of time (~ 21 s). In parallel, the level of exfoliation is estimated by the determination of a melt yield stress. It increases when the feed rate (Q) decreases and the screw speed (N) increases. The effect of these two parameters can be expressed through the ratio Q/N . Figure 20.16 shows that the values of the melt yield stress, and thereby the degree of exfoliation, decrease regularly as a function of Q/N . The best exfoliation is thus obtained at high screw speed and low feed rate. Direct morphological

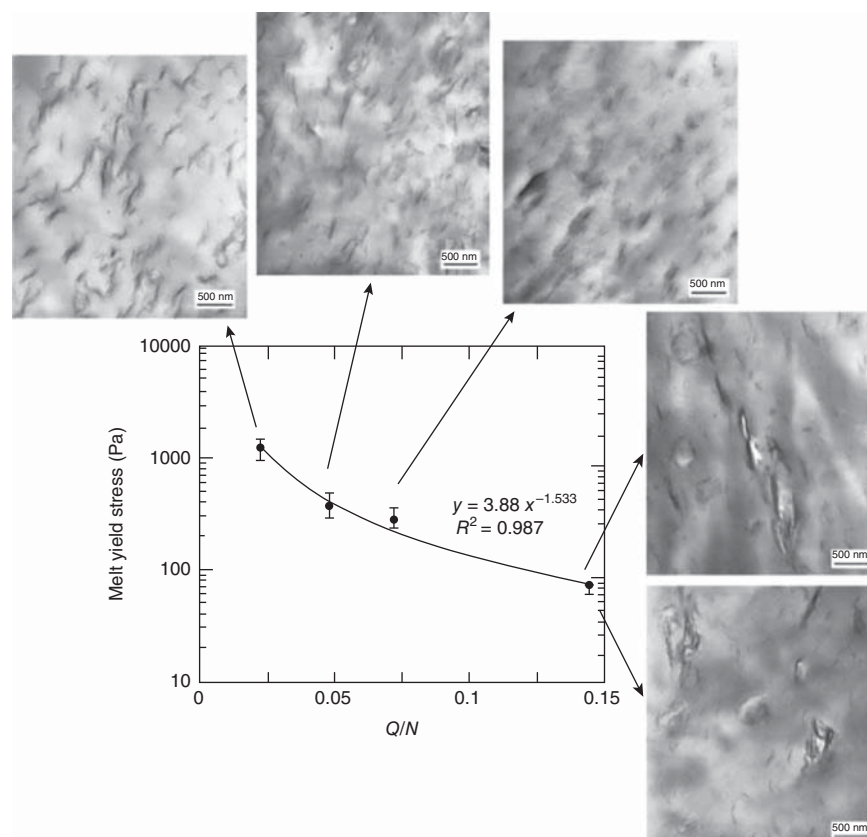


Figure 20.16 Evolution of melt yield stress as a function of Q/N for the nanocomposites prepared with different operating conditions. Lertwimolnun and Vergnes [275]. Reproduced with permission of Elsevier.

observations by TEM confirm the results found by SAOS measurements (rheological measurements in small-amplitude oscillatory shear) [275]. The better dispersion is observed at low Q/N ratio. The difference of morphology is evident between the highest and the lowest Q/N ratios.

20.7 CONCLUDING REMARKS

There are several approaches for developing high-performance PNCs. The critical challenge is the development of methods to improve the dispersion of nanoparticles in a polymer matrix because their enhanced dispersion in polymer matrices greatly improves the mechanical, electrical, and optical properties of nanocomposites. Despite various methods, such as melt processing, solution processing, *in situ* polymerization, sol-gel technology and chemical functionalization, there are still opportunities and challenges to be found in order to improve dispersion and modify interfacial properties. In particular, it is needed to identify stable procedures for homogeneous dispersion of nanoparticles, by directing the structure and properties of the polymer matrix during crystallization/solidification taking into consideration the materials' characteristics, the nanoparticles' shape, as well as the conditions of the processing.

REFERENCES

1. Fukushima Y, Okada A, Kawasumi M, Kurauchi T, Kamigaito O. Swelling behavior of montmorillonite by poly-6-amide. *Clay Miner* 1998;23:27–34.
2. Vaia RA, Ishii H, Giannelis EP. Synthesis and properties of two-dimensional nanostructures by direct intercalation of polymer melts in layered silicates. *Chem Mater* 1993;5:1694–1696.
3. Lan T, Pinnavaia TJ. Clay-reinforced epoxy nanocomposites. *Chem Mater* 1994;6:2216–2219.
4. Winey K, Vaia R, editors. *Polymer Nanocomposites*. Pittsburgh, PA: MRS Bulletin; Materials Research Society; 2007.
5. Nanocomposites (Carbon Nanotubes, Polymer Metal Fiber, Nanofibers, Graphene, Nanoplatelet and Others) Market for Automotive, Aviation, Electronics, Energy, Construction, Healthcare, Plastics, Military, Consumer Goods and Other Applications - Global Industry Analysis, Size, Share, Growth, Trends and Forecast, 2012–2018 Transparency Market design Albany, New York (PRWEB) August 09, 2013.
6. Giannelis EP. Polymer layered silicate nanocomposites. *Adv Mater* 1996;8:29–35.
7. Dubois MAP. Polymer-layered silicate nanocomposites: preparation, properties and uses of a new class of materials. *Mater Sci Eng R* 2000;R28:1–65.
8. Ray SS, Okamoto M. Polymer/layered silicate nanocomposites: A review from preparation to processing. *Prog Polym Sci* 2003;28:1539–1641.
9. Okamoto M. In: Nalwa HS, editor. *Encyclopedia of Nanoscience and Nanotechnology*. Volume 8. Stevenson Ranch, CA: American Scientific; 2004. p 1–52.
10. Thostenson ET, Li C, Chou TW. Nanocomposites in context. *Compos Sci Technol* 2005;65:491–516.
11. Drummy LF, Koerner H, Farmer BL, Vaia RA. *Advanced Morphology Characterization of Clay-Based Polymer Nanocomposites*, CMS Workshop Lecture Series, Vol. 14. Clay Minerals Society; 2006.
12. Tjong SC. Structural and mechanical properties of polymer nanocomposites. *Mater Sci Eng R* 2006;53:73–197.
13. Hussain F, Hojjati M, Okamoto M, Gorga RE. Review article: Polymer-matrix nanocomposites, processing, manufacturing, and application: An overview. *J Compos Mater* 2006;40(17): 1511–1575.
14. Pavlidou S, Papaspyrides CD. A review on polymer-layered silicate nanocomposites. *Prog Polym Sci* 2008;33:1119–1198.
15. Kiliaris P, Papaspyrides CD. Polymer/layered silicate (clay) nanocomposites: An overview of flame retardancy. *Prog Polym Sci* 2010;35:902–958.
16. Bitinis N, Hernandez M, Verdejo R, Kenny JM, Lopez-Manchado MA. Recent advances in clay/polymer nanocomposites. *Adv Mater* 2011;23:5229–5236.
17. Xie XL, Maia YW, Zhou XP. Dispersion and alignment of carbon nanotubes in polymer matrix: A review. *Mater Sci Eng R* 2005;49:89–112.
18. Moniruzzaman M, Winey KI. Polymer nanocomposites containing carbon nanotubes. *Macromolecules* 2006;39:5194–5205.
19. Sahoo NG, Ranab S, Chob JW, Li L, Chana SH. Polymer nanocomposites based on functionalized carbon nanotubes. *Prog Polym Sci* 2010;35:837–867.
20. Spitalskya Z, Tasisb D, Papagelisb K, Galiotis C. Carbon nanotube-polymer composites: Chemistry, processing, mechanical and electrical properties. *Prog Polym Sci* 2010; 35:357–401.
21. Domènech B, Bastos-Arrieta J, Alonso A, Muñoz M, Muraviev DN, Macanás J. Bifunctional polymer-metal nanocomposite ion exchange materials. In: Kilislioglu A, editor. *Ion Exchange Technologies*. InTech 2012, chapter 3. p 35–72.
22. Silvestre C, Duraccio D, Cimmino S. Food packaging based on polymer nanomaterials. *Prog Polym Sci* 2011;36:1766–1782.
23. Silvestre C, Cimmino S, editors. *Ecosustainable Polymer Nanomaterials for Food Packaging*. Boca Raton, FL: CRC Press Taylor and Francis Group; 2013.
24. Pinnavaia TJ, Beall GWJ. *Polymer-Clay Nanocomposites*. New York: John Wiley & Sons; 2001.
25. Krishnamoorti R, Vaia RA, editors. *Polymer Nanocomposites: Synthesis, Characterization, and Modeling*; ACS Symposium Series. Washington, DC: American Chemical Society; 2001.
26. Ray SS, Bousmina M. *Polymer Nanocomposites and Their Applications*. Stevenson Ranch, CA: American Scientific; 2006.
27. Mai YW, Yu ZZ, editors. *Polymer Nanocomposites*. Boca Raton, FL: CRC; 2006.
28. Wilkie CA, Morgan AB, editors. *Fire Retardancy of Polymeric Materials*. 2nd ed. Boca Raton, FL, USA: CRC Press; 2009. ISBN 9781420083996.
29. Ruckenstein E, Li ZF. Surface modification and functionalization through the self-assembled monolayer and graft polymerization. *Adv Colloid Interface Sci* 2005;113:43–63.

30. Miranda-Trevino JC, Coles CA. Kaolinite properties, structure and influence of metal retention on pH. *Appl Clay Sci* 2003;23:133–139.
31. Beyer G. Nanocomposites: A new class of flame retardants for polymers. *Plast Addit Compd* 2002;4:22–27.
32. McNally T, Murphy WR, Lew CY, Turner RJ, Brennan GP. Polyamide12-layered silicate nanocomposites by melt compounding. *Polymer* 2003;44:2761–2772.
33. Solomon MJ, Almusallam AS, Seefeldt KF, Somwangthanaroj A, Varadan P. Rheology of polypropylene/clay hybrid materials. *Macromolecules* 2001;34:1864–1872.
34. Manias E, Touny A, Wu L, Strawhecker K, Lu B, Chung TC. Polypropylene/ montmorillonite nanocomposites. Review of the synthetic routes and materials properties. *Chem Mater* 2001;13:3516–3523.
35. Hedley CB, Yuan G, Theng BKG. Thermal analysis of montmorillonites modified with quaternary phosphonium and ammonium surfactants. *Appl Clay Sci* 2007;35:180–188.
36. Kornmann X, Lindberg H, Berglund LA. Synthesis of epoxy–clay nanocomposites: influence of the nature of the clay on structure. *Polymer* 2001;42:1303–1310.
37. Zanetti M, Lomakin S, Camino G. Polymer layered silicate nanocomposites. *Macromol Mater Eng* 2000;279:1–9.
38. Xie W, Gao Z, Liu K, Pan WP, Vaia R, Hunter D, et al. Thermal characterization of organically modified montmorillonite. *Thermochim Acta* 2001;367(368):339–350.
39. Kornmann X, Lindberg H, Berglund LA. Synthesis of epoxy–clay nanocomposites: influence of the nature of the clay on structure. *Polymer* 2001;42:1303–1310.
40. Kim CM, Lee DH, Hoffmann B, Kressler J, Stoppelmann G. Influence of nanofillers on the deformation process in layered silicate/polyamide12 nanocomposites. *Polymer* 2001;42:1095–1100.
41. Zerda AS, Lesser AJ. Intercalated clay nanocomposites: morphology, mechanics and fracture behavior. *J Polym Sci Polym Phys* 2001;39:1137–1146.
42. Lan T, Kaviratna PD, Pinnavia TJ. Mechanism of clay tactoid exfoliation in epoxy-clay nanocomposites. *Chem Mater* 1995;7:2144–2150.
43. Pandey JK, Reddy KR, Kumar AP, Singh RP. An overview of the degradability of polymer nanocomposites. *Polym Degrad Stab* 2005;88:234–250.
44. Miyagawa H, Drzal LT, Miyagawa H, Carsello JA. Intercalation and exfoliation of clay nanoplatelets in epoxy-based nanocomposites: TEM and XRD observations. *Polym Eng Sci* 2006;46:453–463.
45. Iijima S. Helical microtubules of graphitic carbon. *Nature* 1991;354:56–58.
46. Ajayan PM, Stephan O, Colliex C, Trauth D. Aligned carbon nanotube arrays formed by cutting a polymer resin—nanotube composite. *Science* 1994;265:1212–1214.
47. Cooper CA, Young RJ, Halsall M. Investigation into the deformation of carbon nanotubes and their composites through the use of Raman spectroscopy. *Composites A* 2001;32:401–411.
48. Gao G, Cagin T, Goddard WA III. Energetics, structure, mechanical and vibrational properties of single-walled carbon nanotubes. *Nanotechnology* 1998;9:184–191.
49. Uchida T, Kumar S. Single wall carbon nanotube dispersion and exfoliation in polymers. *J Appl Polym Sci* 2005;98:985–989.
50. De Heer Walt A. Nanotubes and the pursuit of applications. *MRS Bull* 2004;29:281–285.
51. Dai H. Carbon nanotubes: Synthesis, integration, and properties. *Acc Chem Res* 2002;35:1035–1044.
52. Awasthi K, Srivastava A, Srivastava ON. Synthesis of carbon nanotubes. *J Nanosci Nanotechnol* 2005;5:1616–1636.
53. Monthieux M. Filling single-wall carbon nanotubes. *Carbon* 2002;40:1809–1823.
54. Thostenson ET, Zhifeng R, Chou TW. Advances in the science and technology of carbon nanotubes and their composites: A review. *Compos Sci Technol* 2001;61:1899–1912.
55. Dyke CA, Tour JM. Overcoming the insolubility of carbon nanotubes through high degree of sidewall functionalization. *Chem Eur J* 2004;10:812–817.
56. Haddon RC. Solution properties of single-walled carbon nanotubes. *Science* 1998;282:95–98.
57. Michelson ET, Huffman CB, Rinzler AG, Smalley RE, Hauge RH, Margrave JL. Fluorination of single-wall carbon nanotubes. *Chem Phys Lett* 1998;296:188–194.
58. Michelson ET, Chiang IW, Zimmerman JL, Boul PJ, Lozano J, Liu J, Smalley RE, Hauge RH, Margrave JL. Solvation of fluorinated single-wall carbon nanotubes in alcohol solvents. *J Phys Chem B* 1999;103:4318–4322.
59. Chen J, Rao AM, Lyuksyutov S, Itkis ME, Hamon MA, Hu H, Cohn RW, Eklund PC, Colbert DT, Smalley RE, Haddon RC. Dissolution of full-length single-walled carbon nanotubes. *J Phys Chem B* 2001;105:2525–2528.
60. Riggs JE, Guo Z, Carroll DL, Sun YP. Strong luminescence of solubilized carbon nanotubes. *J Am Chem Soc* 2000;122:5879–5880.
61. Riggs JE, Walker DB, Carroll DL, Sun Y-P. Optical limiting properties of suspended and solubilized carbon nanotubes. *J Phys Chem B* 2000;104:7071–7076.
62. Jin Z, Sun X, Xu G, Goh SH, Ji W. Nonlinear optical properties of some polymer/multiwalled carbon nanotube composites. *Chem Phys Lett* 2000;318:505–510.
63. Niyogi S, Hu H, Hamon MA, Bhowmik P, Zhao B, Rozenzhak SM, Chen J, Itkis ME, Meier MS, Haddon RC. Chromatographic purification of soluble single-walled carbon nanotubes (s-SWNTs). *J Am Chem Soc* 2001;123:733–734.
64. Zhao B, Hu H, Niyogi S, Itkis ME, Hamon MA, Bhowmik P, Meier MS, Haddon RC. Chromatographic purification and properties of soluble single-walled carbon nanotubes. *J Am Chem Soc* 2001;123:11673–11677.
65. Sun YP, Huang W, Lin Y, Fu K, Kitaygorodskiy A, Riddle LA, Yu YJ, Carroll DL. Soluble dendron-functionalized carbon nanotubes: Preparation, characterization, and properties. *Chem Mater* 2001;13:2864–2869.
66. Fu K, Huang W, Lin Y, Riddle LA, Carroll DL, Sun YP. Defunctionalization of functionalized carbon nanotubes. *Nano Lett* 2001;1:439–441.
67. Sun Y, Wilson SR, Schuster DI. High dissolution and strong light emission of carbon nanotubes in aromatic amine solvents. *J Am Chem Soc* 2001;123:5348–5349.

68. Bahr JL, Yang J, Kosynkin DV, Bronikowski MJ, Smalley RE, Tour JM. Functionalization of carbon nanotubes by electrochemical reduction of aryl diazonium salts: A bucky paper electrode. *J Am Chem Soc* 2001;123:6536–6542.
69. Bahr JL, Tour JM. Highly functionalized carbon nanotubes using in situ generated diazonium compounds. *Chem Mater* 2001;13:3823–3824.
70. Pekker S, Salvatat JP, Jakab E, Bonard JM, Forro L. Hydrogenation of carbon nanotubes and graphite in liquid ammonia. *J Phys Chem B* 2001;105:7938–7943.
71. Czerw R, Guo Z, Ajayan PM, Sun YP, Carroll DL. Organization of polymers onto carbon nanotubes: A route to nanoscale assembly. *Nano Lett* 2001;1:423–427.
72. Lin Y, Rao AM, Sadanadan B, Kenik EA, Sun YP. Functionalizing multiple-walled carbon nanotubes with aminopolymers. *J Phys Chem B* 2002;106:1294–1298.
73. Huang W, Lin Y, Taylor S, Gaillard J, Rao AM, Sun YP. Sonication-assisted functionalization and solubilization of carbon nanotubes. *Nano Lett* 2002;2:231–234.
74. Holzinger M, Vostrowsky O, Hirsch A, Hennrich F, Kappes M, Weiss R, Jellen F. Sidewall functionalization of carbon nanotubes. *Angew Chem Int Ed* 2001;40:4002–4005.
75. Georgakilas V, Kordatos K, Prato M, Guldi DM, Holzinger M, Hirsch A. Organic functionalization of carbon nanotubes. *J Am Chem Soc* 2002;120:760–761.
76. Liu J, Rinzler AG, Dai H, Hafner JH, Bradley RK, Boul PJ, Lu A, Iverson T, Shelimov K, Huffman CB, Rodriguez-Macias F, Shon YS, Lee TR, Colbert DT, Smalley RE. Fullerene pipes. *Science* 1998;280:1253–1256.
77. Hamon MA, Hu H, Bhowmik P, Niyogi S, Zhao B, Itkis ME, Haddon RC. End-group and defect analysis of soluble single-walled carbon nanotubes. *Chem Phys Lett* 2001;347:8–12.
78. Behra R, Sigg L, Clift MJD, Herzog F, Minghetti M, Johnston B, Petri-Fink A, Rothen-Rutishauser B. Bioavailability of silver nanoparticles and ions: from a chemical and biochemical perspective. *J R Soc Interface* 2013;10.
79. Klasen H. Historical review of the use of silver in the treatment of burns. I. Early uses. *Burns* 2000;26:117–130.
80. Maillard JY, Hartemann P. Silver as an antimicrobial: Facts and gaps in knowledge. *Crit Rev Microbiol* 2012;39:373–383.
81. Rai M, Yadav A, Gade A. Silver nanoparticles as a new generation of antimicrobials. *Biotechnol Adv* 2009;27:76–83.
82. Eustis S, Krylova G, Smirnova N, Eremenko A, Tabor C, Huang W, El-Sayed MA. Using silica films and powders modified with benzophenone to photoreduce silver nanoparticles. *J Photochem Photobiol A* 2006;181:385–393.
83. Van Hoonacker A, Englebienne P. Revisiting silver nanoparticle chemical synthesis and stability by optical spectroscopy. *Curr Nanosci* 2006;2:359–371.
84. Pastoriza-Santos I, Liz-Marzan LM. N,N-dimethylformamide as a reaction medium for metal nanoparticle synthesis. *Adv Funct Mater* 2009;19:679–688.
85. Shirtcliffe N, Nickel U, Schneider S. Reproducible preparation of silver sols with small particle size using borohydride reduction: for use as nuclei for preparation of larger particles. *J Colloid Interface Sci* 1999;211:122–129.
86. Nickel U, Castell AZ, Poppl K, Schneider S. A silver colloid produced by reduction with hydrazine as support for highly sensitive surface-enhanced Raman. *Langmuir* 2000;16:9087–9091.
87. Fievet F, Francoise FV, Lagier JP, Dumont B, Figlarz M. Controlled nucleation and growth of micrometre-size copper particles prepared by the polyol process. *J Mater Chem* 1993;3:627–632.
88. Leopold N, Lendl B. A new method for fast preparation of highly surface-enhanced Raman scattering (SERS) active silver colloids at room temperature by reduction of silver nitrate with hydroxylamine hydrochloride. *J Phys Chem B* 2003;107:5723–5727.
89. Zhang Y, Peng H, Huang W, Zhou Y, Zhang X, Yan D. Hyperbranched poly(amidoamine) as the stabilizer and reductant to prepare colloid silver nanoparticles in situ and their antibacterial activity. *J Phys Chem C* 2008;112:2330–2336.
90. Yin Y, Li ZY, Zhong Z, Gates B, Xia Y, Venkateswaran S. Synthesis and characterization of stable aqueous dispersions of silver nanoparticles through the Tollens process. *J Mater Chem* 2002;12:522–527.
91. Liu YC, Lin LH. New pathway for the synthesis of ultrafine silver nanoparticles from bulk silver substrates in aqueous solutions by sonoelectrochemical methods. *Electrochem Commun* 2004;6:1163–1168.
92. Raveendran P, Fu J, Wallen SL. Completely ‘green’ synthesis and stabilization of metal nanoparticles. *J Am Chem Soc* 2003;125:940–941.
93. Cobley CM, Rycenga M, Zhou F, Li ZY, Xia Y. Etching and growth: An intertwined pathway to silver nanocrystals with exotic shapes. *Angew Chem Int Ed* 2009;48:4824–4827.
94. Sun Y, Xia Y. Shape-controlled synthesis of gold and silver nanoparticles. *Science* 2002;298:2176–2179.
95. Sharma VK, Yngard RA, Lin Y. Silver nanoparticles: Green synthesis and their antimicrobial activities. *Adv Colloid Interface Sci* 2009;145:83–96.
96. Vallet-Regi M, Balas F, Arcos D. Mesoporous materials for drug delivery. *Angew Chem Int Ed* 2007;46:7548–7558.
97. Ghorai TK, Dhak D, Biswas SK, Dalai S, Pramanik P. Photocatalytic oxidation of organic dyes by nano-sized metal molybdate incorporated titanium dioxide ($M_xMo_xTi_{1-x}O_6$) ($M = Ni, Cu, Zn$) photocatalysts. *J Mol Catal A* 2007;273:224–229.
98. Rossignol C, Verelst M, Dexpert-Ghys J, Rul S. Synthesis of undoped ZnO nanoparticles by spray pyrolysis. *Adv Sci Technol* 2006;45:237–241.
99. Babelon P, Dequidt AS, Mostefa-Sba H, Bourgeois S, Sibillot P, Sacilotti M. SEM and XPS studies of titanium dioxide thin films grown by MOCVD. *Thin Solid Films* 1998;322:63–67.
100. Kim BH, Lee JY, Choa YH, Higuchi M, Mizutani N. Preparation of TiO_2 thin film by liquid sprayed mist CVD method. *Mater Sci Eng B* 2004;107:289–294.
101. Okuyama K, Shimada M, Fujimoto T, Maekawa T, Nakaso K, Seto T. Effects of preparation conditions on the characteristics of titanium dioxide particles produced by a CVD method. *J Aerosol Sci* 1998;29(1):S907–S908.
102. Gu DE, Yang BC, Hu YD. V and N co-doped nanocrystal anatase TiO_2 photocatalysts with enhanced photocatalytic activity under visible light irradiation. *Catal Commun* 2008;9:1472–1476.

103. Zhang X, Liu Q. Preparation and characterization of titania photocatalyst co-doped with boron, nickel, and cerium. *Mater Lett* 2008;62:2589–2592.
104. Peng F, Cai L, Huang L, Yu H, Wang H. Preparation of nitrogen-doped titanium dioxide with visible-light photocatalytic activity using a facile hydrothermal method. *J Phys Chem Solids* 2008;69(7):1657–1664.
105. Peng F, Cai L, Huang L, Yu H, Wang H. Synthesis and characterization of substitutional and interstitial nitrogen-doped titanium dioxides with visible light photocatalytic activity. *J Solid State Chem* 2008;181:130–136.
106. Feng X, Wang Q, Wang G, Qiu F. Preparation of Nano-TiO₂ by ethanol-thermal method and its catalytic performance for synthesis of dibutyl carbonate by transesterification. *Chin J Catal* 2006;27(3):195–196.
107. Zou H, Wu S, Shen J. Polymer/silica nanocomposites: Preparation, characterization, properties, and applications. *Chem Rev* 2008;108:3893–3957.
108. Rong MZ, Zhang MQ, Zheng YX, Zeng HM, Walter R, Friedrich K. Structure–property relationships of irradiation grafted nano-inorganic particle filled polypropylene composites. *Polymer* 2001;42:167–183.
109. Neouze MA, Schubert U. Surface modification and functionalization of metal and metal oxide nanoparticles by organic ligands. *Monatsh Chem* 2008;139:183–195.
110. Kickelbick G, Schubert U. Organic functionalization of metal oxide nanoparticles. In: Baraton MI, editor. *Synthesis, Functionalization and Surface Treatment of Nanoparticles*. Stevenson Ranch; 2003. p 91.
111. Doty RC, Tshikhudo TR, Brust M, Fernig DG. Extremely stable water-soluble Ag nanoparticles. *Chem Mater* 2005;17:4630–4637.
112. Schulz-Dobrick M, Sarathy KV, Jansen M. Surfactant-free synthesis and functionalization of gold nanoparticles. *J Am Chem Soc* 2005;127:12816–12817.
113. Warner MG, Hutchison JE. Synthesis and assembly of functionalized gold nanoparticles. In: Baraton MI, editor. *Synthesis, Functionalization and Surface Treatment of Nanoparticles*. Stevenson Ranch; 2003. p 67.
114. Roux S, Garcia B, Bridot JL, Salome M, Marquette C, Lemelle L, Gillet P, Blum L, Perriat P, Tillement O. Synthesis, characterization of dihydrolipoic acid capped gold nanoparticles, and functionalization by the electroluminescent luminol. *Langmuir* 2005;21:2526–2536.
115. Zayats M, Katz E, Baron R, Willner I. Reconstitution of apo-glucose dehydrogenase on pyrroloquinoline quinone-functionalized Au nanoparticles yields an electrically contacted biocatalyst. *J Am Chem Soc* 2005;127:12400–12406.
116. Ulman A. Formation and structure of self-assembled monolayers. *Chem Rev* 1996;96:1533–1534.
117. Fan H, Chen Z, Brinker CJ, Clawson J, Alam T. Synthesis of organo-silane functionalized nanocrystal micelles and their self-assembly. *J Am Chem Soc* 2005;127:13746–13747.
118. Ramirez E, Jansat S, Philippot K, Lecante P, Gomez M, Masdeu-Bulto AM, Chaudret B. Influence of organic ligands on the stabilization of palladium nanoparticles. *J Organomet Chem* 2004;689:4601–4610.
119. Woehrle GH, Hutchison JE. Thiol-functionalized undecagold clusters by ligand exchange: Synthesis, mechanism, and properties. *Inorg Chem* 2005;44:6149–6158.
120. Dong TY, Wu HH, Lin MC. Superlattice of octanethiol-protected copper nanoparticles. *Langmuir* 2006;22:6754–6756.
121. Gooding JJ, Mearns F, Yang W, Liu J. Self-assembled monolayers into the 21st century: Recent advances and applications. *Electroanalysis* 2003;15:81–96.
122. Liu F, Quan B, Liu Z, Chen L. Surface characterization study on SnO₂ powder modified by thiourea. *Mater Chem Phys* 2005;93:301–304.
123. Elbhir Z, Chevalier Y, Chovelon JM, Jaffrezic-Renault N. Grafting of phosphonate groups on the silica surface for the elaboration of ion-sensitive field-effect transistors. *Talanta* 2000;52:495–507.
124. Hua F, Swihart MT, Ruckenstein E. Efficient surface grafting of luminescent silicon quantum dots by photoinitiated hydrosilylation. *Langmuir* 2005;21:6054–6062.
125. Rothon RN. *Particulate-Filled Polymer Composites*. Harlow, U.K.: Longman; 1995. p 123.
126. del Campo A, Sen T, Lellouche JP, Bruce IJ. Multifunctional magnetite and silica–magnetite nanoparticles: Synthesis, surface activation and applications in life sciences. *J Magn Magn Mater* 2005;293:33–40.
127. Kikukawa T, Kuraoka K, Kawabe K, Yamashita M, Fukumi K, Hirao K, Yazawa T. Stabilities and pore size effect of proton-conducting organic–inorganic hybrid membranes prepared through surface modification of porous glasses. *J Membr Sci* 2005;259:161–166.
128. Li JJ, Xu XY, Jiang Z, Hao ZP, Hu C. Nanoporous silica-supported nanometric palladium: Synthesis, characterization, and catalytic deep oxidation of benzene. *Environ Sci Tech* 2005;39:1319–1323.
129. Pere E, Cardy H, Latour V, Lacombe S. Low-temperature reaction of trialkoxysilanes on silica gel: A mild and controlled method for modifying silica surfaces. *J Colloid Interface Sci* 2005;281:410–416.
130. Reculosa S, Poncet-Legrand C, Perro A, Duguet E, Bourgeat-Lami E, Mingotaud C, Ravaine S. Hybrid dissymmetrical colloidal particles. *Chem Mater* 2005;17:3338–3344.
131. Wang YP, Pei XW, Yuan K. Reverse ATRP grafting from silica surface to prepare well-defined organic/inorganic hybrid nanocomposite. *Mater Lett* 2004;59:520–523.
132. Wu CG, Tzeng LF, Kuo YT, Shu CH. Enhancement of the photocatalytic activity of TiO₂ film via surface modification of the substrate. *Appl Catal A* 2002;226:199–211.
133. Ruan WH, Mai YL, Wang XH, Rong MZ, Zhang MQ. Effects of processing conditions on properties of nano-SiO₂/polypropylene composites fabricated by pre-drawing technique. *Compos Sci Technol* 2007;67:2747–2756.
134. Lagaly G. Introduction: from clay mineral-polymer interactions to clay mineral-polymer nanocomposites. *Appl Clay Sci* 1999;15:1–9.
135. Greenland DJ. Adsorption of poly(vinyl alcohols) by montmorillonite. *J Colloid Sci* 1963;18:647–664.
136. Ogata N, Kawakage S, Ogihara T. Poly(vinyl alcohol)/clay and poly(ethylene oxide)/clay blend prepared using water as solvent. *J Appl Polym Sci* 1997;66:573–581.

137. Parfitt RL, Greenland DJ. Adsorption of poly(ethylene glycols) on montmorillonites. *Clay Miner* 1970;8:305–315.
138. Ruiz-Hitzky E, Aranda P, Casal B, Galvan JC. Nanocomposite materials with controlled ion mobility. *Adv Mater* 1995;7:180–184.
139. Levy R, Francis CW. Interlayer adsorption of polyvinylpyrrolidone on montmorillonite. *J Colloid Interface Sci* 1975;50:442–450.
140. Billingham J, Breen C, Yarwood J. Adsorption of polyamine, polyacrylic acid and polyethylene glycol on montmorillonite: An in situ study using ATR-FTIR. *Vib Spectrosc* 1997;14:19–34.
141. Zhao X, Urano K, Ogasawara S. Adsorption of poly(ethylene vinyl alcohol) from aqueous solution on montmorillonite clays. *Colloid Polym Sci* 1989;267:899–906.
142. Ogata N, Jimenez G, Kawai H, Ogihara T. Structure and thermal/mechanical properties of poly(l-lactide)–clay blend. *J Polym Sci Polym Phys* 1997;35:389–396.
143. Jeon HG, Jung HT, Lee SW, Hudson SD. Morphology of polymer silicate nanocomposites. High density polyethylene and a nitrile. *Polym Bull* 1998;41:107–113.
144. Jimenez G, Ogata N, Kawai H, Ogihara T. Structure and thermal/mechanical properties of poly(γ -caprolactone)–clay blend. *J Appl Polym Sci* 1997;64:2211–2220.
145. Srivastava SK, Pramanik M, Acharya H. Ethylene/vinyl acetate copolymer/clay nanocomposites. *J Polym Sci Polym Phys* 2006;44:471–480.
146. Tseng CR, Wu JY, Lee HY, Chang FC. Preparation and crystallization behavior of syndiotactic polystyrene–clay nanocomposites. *Polymer* 2001;42:10063–10070.
147. Krikorian V, Pochan D. Poly(l-lactide acid)/layered silicate nanocomposite: Fabrication, characterization, and properties. *Chem Mater* 2003;15:4317–4324.
148. Chang JH, Uk-An Y, Sur GS. Poly(lactic acid) nanocomposites with various organoclays. I. thermomechanical properties, morphology, and gas permeability. *J Polym Sci Polym Phys* 2003;41:94–103.
149. Yano K, Usuki A, Okada A, Kurauchi T, Kamigaito O. Synthesis and properties of polyimide/clay hybrid. *J Polym Sci Polym Chem* 1993;31:2493–2498.
150. Ray SS, Bousima M. Biodegradable polymers and their layered silicate nanocomposites: In greening the 21st century materials world. *Prog Mater Sci* 2005;50:962–1079.
151. Xia H, Wang Q, Qiu G. Polymer-encapsulated carbon nanotubes prepared through ultrasonically initiated in situ emulsion polymerization. *Chem Mater* 2003;15:3879–3886.
152. Suslick KS. Sonochemistry. *Science* 1990;247:1439–1445.
153. Li C, Pang XJ, Qu MZ, Zhang QT, Wang B, Zhang BL, et al. Fabrication and characterization of polycarbonate/carbon nanotubes composites. *Composites A* 2005;37:1485–1489.
154. Safadi B, Andrews R, Grulke EA. Multiwalled carbon nanotubes polymer composites: Synthesis and characterization of thin films. *J Appl Polym Sci* 2002;84:2660–2669.
155. Sahoo NG, Jung YC, Yoo HJ, Cho JW. Effect of functionalized carbon nanotubes on molecular interaction and properties of polyurethane composites. *Macromol Chem Phys* 2006;207:1773–1780.
156. Chen W, Tao X. Self-organizing alignment of carbon nanotubes in thermoplastic polyurethane. *Macromol Rapid Commun* 2005;26:1763–1767.
157. Qian D, Dickey EC, Andrews R, Rantell T. Load transfer and deformation mechanisms in carbon nanotube-polystyrene composites. *Appl Phys Lett* 2000;76:2868–2870.
158. Wong M, Paramsothy M, Xu XJ, Ren Y, Li S, Liao K. Physical interaction at carbon nanotube-polymer interface. *Polymer* 2003;44:7757–7764.
159. Kota AK, Cipriano BH, Duesterberg MK, Gershon AL, Powell D, Raghavan SR, et al. Electrical and rheological percolation in polystyrene/MWCNT nanocomposites. *Macromolecules* 2007;40:7400–7406.
160. Sandler J, Shaffer MSP, Prasse T, Bauhofer W, Schulte K, Windle AH. Development of a dispersion process for carbon nanotubes in an epoxy matrix and the resulting electrical properties. *Polymer* 1999;40:5967–5971.
161. Song YS, Youn JR. Influence of dispersion states of carbon nanotubes on physical properties of epoxy nanocomposites. *Carbon* 2005;43:1378–1385.
162. Shaffer MSP, Windle AH. Fabrication and characterization of carbon nanotube/poly(vinyl alcohol) composites. *Adv Mater* 1999;11:937–941.
163. Chae HG, Sreekumar TV, Uchida T, Kumar S. A comparison of reinforcement efficiency of various types of carbon nanotubes in polyacrylonitrile fiber. *Polymer* 2005;46:10925–10935.
164. Bin Y, Kitanaka M, Zhu D, Matsuo M. Development of highly oriented polyethylene filled with aligned carbon nanotubes by gelation/crystallization from solutions. *Macromolecules* 2003;36:6213–6219.
165. Oberdisse J, Hine P, Pyckhout-Hintzen W. Structure of interacting aggregates of silica nanoparticles in a polymer matrix: small-angle scattering and reverse Monte Carlo simulations. *Soft Matter* 2007;3:476–485.
166. Zhang Q, Archer LA. Optical polarimetry and mechanical rheometry of poly(ethylene oxide)–silica dispersions. *Macromolecules* 2004;37:1928.
167. van Zyl WE, García M, Schrauwen BAG, Kooi BJ, De Hosson JM, Verweij H. Hybrid polyamide/silica nanocomposites: Synthesis and mechanical testing. *Macromol Mater Eng* 2002;287:106–110.
168. Hu N, Zhou H, Dang G, Rao X, Chen C, Zhang W. Efficient dispersion of multi-walled carbon nanotubes by in situ polymerization. *Polym Int* 2007;56:655–659.
169. So HH, Cho JW, Sahoo NG. Effect of carbon nanotubes on mechanical and electrical properties of polyimide/carbon nanotubes nanocomposites. *Eur Polym J* 2007;43:3750–3756.
170. Beyer G. Nanocomposites: A new class of flame retardants for polymers. *Plast Addit Compd* 2002;4(10):22–27.
171. Okamoto M, Morita S, Taguchi H, Kim YH, Kotaka T, Tateyama H. Synthesis and structure of smectic clay/poly(methyl methacrylate) and clay polystyrene nanocomposites via in situ intercalative polymerization. *Polymer* 2000;41:3887–3890.
172. Okamoto M, Morita S, Kotaka T. Dispersed structure and ionic conductivity of smectic clay/polymer nanocomposites. *Polymer* 2001;42:2685–2688.

173. Akelah A, Moet M. Polymer-claynanocomposites: Free radical grafting of polystyrene on to organophilic montmorillonite interlayers. *J Mater Sci* 1996;31:3589–3596.
174. Hsu SLC, Chang KC. Synthesis and properties of polybenzoxazole-clay nanocomposites. *Polymer* 2002;43:4097–4101.
175. Sun T, Garces JM. High-performance polypropylene-clay nanocomposites by in-situ polymerization with metallocene/clay catalysts. *Adv Mater* 2002;14:128–130.
176. Bergman JS, Chen H, Giannelis EP, Thomas MG, Coates GW. Synthesis and characterization of polyolefin-silicate nanocomposites: A catalyst intercalation and in situ polymerization approach. *J Chem Soc Chem Commun* 1999;21:2179–2180.
177. Tudor J, Willington L, O'Hare D, Royan B. Intercalation of catalytically active metal complexes in phyllosilicates and their application as propene polymerization catalyst. *Chem Commun* 1996;17:2031–2032.
178. Jin YH, Park HJ, Im SS, Kwak SY, Kwak S. Polyethylene/clay nanocomposite by in situ exfoliation of montmorillonite during Ziegler-Natta polymerization of ethylene. *Macromol Rapid Commun* 2002;23:135–140.
179. Ke YC, Long CF, Qi ZN. Crystallization, properties, and crystal and nanoscale morphology of PET-clay nanocomposites. *J Appl Polym Sci* 1999;71:1139–1146.
180. Jia Z, Wang Z, Xu C, Liang J, Wei B, Wu D, et al. Study on poly(methyl methacrylate): Carbon nanotube composites. *Mater Sci Eng A* 1999;271:395–400.
181. Putz KW, Mitchell CA, Krishnamoorti R, Green PF. Elastic modulus of single-walled carbon nanotube/poly(methyl methacrylate) nanocomposites. *J Polym Sci B* 2004;42:2286–2293.
182. Sung JH, Kim HS, Jin HJ, Choi HJ, Chin IJ. Nanofibrous membranes prepared by multiwalled carbon nanotube/poly(methyl methacrylate) composites. *Macromolecules* 2004;37:9899–9902.
183. Hwang GL, Shieh YT, Hwang KC. Efficient load transfer to polymer grafted multiwalled carbon nanotubes in polymer composites. *Adv Funct Mater* 2004;14:487–491.
184. Karim MR, Lee CJ, Park YT, Lee MS. SWNTs coated by conducting polyaniline: Synthesis and modified properties. *Synth Met* 2005;151:131–135.
185. Long Y, Chen Z, Zhang X, Zhang J, Liu Z. Electrical properties of multiwalled carbon nanotube/polypyrrole nanocables: Percolation dominated conductivity. *J Phys D Appl Phys* 2004;37:1965–1969.
186. An KH, Jeong SY, Hwang HR, Lee YH. Enhanced sensitivity of a gas sensor incorporating single-walled carbon nanotube-polypyrrole nanocomposites. *Adv Mater* 2004;16:1005–1009.
187. Fan J, Wan M, Zhu D, Chang B, Pan Z, Xie S. Synthesis, characterizations, and physical properties of carbon nanotubes coated by conducting polypyrrole. *J Appl Polym Sci* 1999;74:2605–2610.
188. Yoo HJ, Jung YC, Sahoo NG, Cho JW. Electroactive shape memory polyurethane nanocomposites from in-situ polymerization with carbon nanotubes. *J Macromol Sci B* 2006;45:441–451.
189. Gao J, Itkis ME, Yu A, Bekyarova E, Zhao B, Haddon RC. Continuous spinning of a single-walled carbon nanotube-nylon composite fiber. *J Am Chem Soc* 2005;127:3847–3854.
190. Jung YC, Sahoo NG, Cho JW. Polymeric nanocomposites of polyurethane block copolymers and functionalized multi-walled carbon nanotubes as crosslinkers. *Macromol Rapid Commun* 2006;27:126–131.
191. Sahoo NG, Jung YC, So HH, Cho JW. Polypyrrole coated carbon nanotubes: synthesis, characterization, and enhanced electrical properties. *Synth Met* 2007;157:374–379.
192. Andrews R, Jacques D, Minot M, Rantell T. Fabrication of carbon multiwall nanotube/polymer composites by shear mixing. *Macromol Mater Eng* 2002;287:395–403.
193. Cho JW, Paul DR. Nylon 6 nanocomposites by melt compounding. *Polymer* 2001;42:1083–1094.
194. Vaia RA, Giannelis EP. Polymer melt intercalation in organically modified layered silicates: model predictions and experiment. *Macromolecules* 1997;30:8000–8009.
195. Fornes TD, Yoon PJ, Keskkula H, Paul DR. Nylon 6 nanocomposites: the effect of matrix molecular weight. *Polymer* 2011;42:9929–9940.
196. Huang JC, Zhu ZK, Yin J, Qian XF, Sun YY. Poly(etherimide)/montmorillonite nanocomposites prepared by melt intercalation: morphology, solvent resistance properties and thermal properties. *Polymer* 2001;42:873–877.
197. Fornes TD, Yoon PJ, Hunter DL, Keskkula H, Paul DR. Effect of organoclay structure on nylon 6 nanocomposite morphology and properties. *Polymer* 2002;43:5915–5933.
198. Phang IY, Liu T, Mohamed A, Pramoda KP, Chen L, Shen L, et al. Morphology, thermal and mechanical properties of nylon 12/organoclay nanocomposites prepared by melt compounding. *Polym Int* 2005;54:456–464.
199. Davis CH, Mathias LJ, Gilman JW, Schiraldi DA, Shields JR, Trulove P, et al. Effects of melt-processing conditions on the quality of poly(ethylene terephthalate) montmorillonite clay nanocomposites. *J Polym Sci Polym Phys* 2002;40:2661–2666.
200. Pegoretti A, Kolarik J, Peronic C, Migliaresi C. Recycled poly(ethylene terephthalate) layered silicate nanocomposites: morphology and tensile mechanical properties. *Polymer* 2004;45:2751–2759.
201. Srivastava SK, Pramanik M, Acharya H. Ethylene/vinyl acetate copolymer/clay nanocomposites. *J Polym Sci Polym Phys* 2006;44:471–480.
202. Tang Y, Hu Y, Wang J, Zong R, Gui Z, Chen Z, et al. Influence of organophilic clay and preparation methods on EVA/montmorillonite nanocomposites. *J Appl Polym Sci* 2004;91:2416–2421.
203. Finnigan B, Martin D, Halley P, Truss R, Campell K. Morphology and properties of thermoplastic polyurethane nanocomposites incorporating hydrophilic layered silicates. *Polymer* 2004;45:2249–2260.
204. Shin SYA, Simon LC, Soares JBP, Scholz G. Polyethylene-clay hybrid nanocomposites: in situ polymerization using bifunctional organic modifiers. *Polymer* 2003;44:5317–5321.
205. Gopakumar TG, Lee JA, Kontopoulou M, Parent JS. Influence of clay exfoliation on the physical properties of montmorillonite/polyethylene composites. *Polymer* 2002;43:5483–5491.

206. Thellen C, Orroth C, Froio D, Ziegler D, Lucciarini J, Farrell R, et al. Influence of montmorillonite layered silicate on plasticized poly(lactide) blown films. *Polymer* 2005;46:11716–11727.
207. Ray SS, Maiti P, Okamoto M, Yamada K, Ueda K. New polylactide/layered silicate nanocomposites. 1. Preparation, characterization and properties. *Macromolecules* 2002;35:3104–3110.
208. Pluta M. Morphology and properties of polylactide modified by thermal treatment, filling with layered silicates and plasticization. *Polymer* 2004;45:8239–8251.
209. Di Y, Iannace S, Maio ED, Nicolais L. Nanocomposites by melt intercalation based on polycaprolactone and organoclay. *J Polym Sci Polym Phys* 2003;41:670–678.
210. Gorraasi G, Tortora M, Vittoria V, Pollet E, Lepoittevin B, Alexandre M, et al. Vapor barrier properties of polycaprolactone montmorillonite nanocomposites: effect of clay dispersion. *Polymer* 2003;44:2271–2279.
211. Lopez Manchado MA, Valentini L, Biagiotti J, Kenny JM. Thermal and mechanical properties of single-walled carbon nanotubes polypropylene composites prepared by melt processing. *Carbon* 2005;43:1499–1505.
212. Valentini L, Biagiotti J, Kenny JM, Santucci S. Morphological characterization of single-wall carbon nanotube/polypropylene composites. *Compos Sci Technol* 2003;63:1149–1153.
213. Zhang H, Zhang Z. Impact behaviour of polypropylene filled with multi-walled carbon nanotubes. *Eur Polym J* 2007;43:3197–3207.
214. Tang W, Santare MH, Advani SG. Melt processing and mechanical property characterization of multi-walled carbon nanotube/high density polyethylene (MWNT/HDPE) composite films. *Carbon* 2003;41:2779–2785.
215. Bocchini S, Frache A, Camino G, Claes M. Polyethylene thermal oxidative stabilisation in carbon nanotubes based nanocomposites. *Eur Polym J* 2007;43:3222–3235.
216. McNally T, Potschke P, Halley P, Murphy M, Martin D, Bell SEJ, et al. Polyethylene multiwalled carbon nanotube composites. *Polymer* 2005;46:8222–8232.
217. Potschke P, Fornes TD, Paul DR. Rheological behavior of multiwalled carbon nanotube/polycarbonate composites. *Polymer* 2002;43:3247–3255.
218. Potschke P, Bhattacharyya AR, Janke A. Morphology and electrical resistivity of melt mixed blends of polyethylene and carbon nanotube filled polycarbonate. *Polymer* 2003;44:8061–8069.
219. Potschke P, Bhattacharyya AR, Janke A. Melt mixing of polycarbonate with multiwalled carbon nanotubes: Microscopic studies on the state of dispersion. *Eur Polym J* 2004;40:137–148.
220. Jin Z, Pramoda KP, Xu G, Goh SH. Dynamic mechanical behavior of melt-processed multi-walled carbon nanotube/poly(methyl methacrylate) composites. *Chem Phys Lett* 2001;337:43–47.
221. Zeng J, Saltysiak B, Johnson WS, Schiraldi DA, Kumar S. Processing and properties of poly(methyl methacrylate)/carbon nano fiber composites. *Composites B* 2004;35:173–178.
222. Haggenueller R, Gommans HH, Rinzler AG, Fischer JE, Winey KI. Aligned single-wall carbon nanotubes in composites by melt processing methods. *Chem Phys Lett* 2000;330:219–225.
223. Gorga RE, Cohen RE. Toughness enhancements in poly(methyl methacrylate) by addition of oriented multiwall carbon nanotubes. *J Polym Sci B* 2004;42:2690–2702.
224. Zeng Y, Ying Z, Du J, Cheng HM. Effects of carbon nanotubes on processing stability of polyoxymethylene in melt-mixing process. *J Phys Chem C* 2007;111:13945–13950.
225. Siochi EJ, Working DC, Park C, Lillehei PT, Rouse JH, Topping CC, et al. Melt processing of SWNT-polyimide nanocomposite fibres. *Composites B* 2004;35:439–446.
226. Zhang WD, Shen L, Phang IY, Liu T. Carbon nanotubes reinforced nylon-6 composite prepared by simple melt-compounding. *Macromolecules* 2004;37:256–259.
227. Meincke O, Kaempfer D, Weickmann H, Friedrich C, Vathauer M, Warth H. Mechanical properties and electrical conductivity of carbon-nanotube filled polyamide-6 and its blends with acrylonitrile/ butadiene/styrene. *Polymer* 2004;45:739–748.
228. Wu CL, Zhang MQ, Rong MZ, Friedrich K. Silica nanoparticles filled polypropylene: effects of particle surface treatment, matrix ductility and particle species on mechanical performance of the composite. *Compos Sci Technol* 2005;65:635–645.
229. Reddy CS, Das CK, Narkis M. Propylene-ethylene copolymer nanocomposites: Epoxy resin grafted nanosilica as a reinforcing filler. *Polym Compos* 2005;26:806–812.
230. Gao XW, Hu GJ, Qian ZZ, Ding YF, Zhang SM, Wang DJ, Yang MS. Immobilization of antioxidant on nanosilica and the antioxidative behavior in low density polyethylene. *Polymer* 2007;48:7309–7315.
231. Lee JA, Kontopoulou M, Parent JS. Synthesis and characterization of polyethylene-based ionomer nanocomposites. *Polymer* 2005;46:5040–5049.
232. Tanahashi M, Hirose M, Lee JC, Takeda K. Organic/inorganic nanocomposites prepared by mechanical smashing of agglomerated silica ultrafine particles in molten thermoplastic resin. *Polym Adv Technol* 2006;17:981–990.
233. Kontou E, Anthoulis G. The effect of silica nanoparticles on the thermomechanical properties of polystyrene. *J Appl Polym Sci* 2007;105:1723–1731.
234. Yang F, Nelson GL. Polymer/silica nanocomposites prepared via extrusion. *Polym Adv Technol* 2006;17:320–326.
235. Katsikis N, Zahradnik F, Helmschrott A, Munstedt H, Vital A. Thermal stability of poly(methyl methacrylate)/silica nano- and microcomposites as investigated by dynamic-mechanical experiments. *Polym Degrad Stab* 2007;92:1966–1976.
236. Nodera A, Kanai T. Flame retardancy of polycarbonate–polydimethylsiloxane block copolymer/silica nanocomposites. *J Appl Polym Sci* 2006;101:3862–3868.
237. Ahn SH, Kim SH, Lee SG. Surface modified silica nanoparticle-reinforced poly (ethylene 2,6-naphthalate). *J Appl Polym Sci* 2004;94:812–818.
238. Tanahashi M, Hirose M, Watanabe Y, Lee JC, Takeda K. Silica/perfluoropolymer nanocomposites fabricated by direct melt-compounding: a novel method without surface modification on nano-silica. *J Nanosci Nanotechnol* 2007;7:2433–2442.
239. Chung SC, Hahn WG, Im SS, Oh SG. Poly(ethylene terephthalate)(PET) nanocomposites filled with fumed silicas by melt compounding. *Macromol Res* 2002;10:221–229.

240. Bikiaris D, Karavelidis V, Karayannidis G. A new approach to prepare poly(ethylene terephthalate)/silica nanocomposites with increased molecular weight and fully adjustable branching or crosslinking by SSP. *Macromol Rapid Commun* 2006;27:1199–1205.
241. Todorov LV, Viana JC. Characterization of PET nanocomposites produced by different melt-based production methods. *J Appl Polym Sci* 2007;106:1659–1669.
242. Zhang H, Zhang Z, Yang JL, Friedrich K. Temperature dependence of crack initiation fracture toughness of various nanoparticles filled polyamide 66. *Polymer* 2006;47:679–689.
243. Bogoslovov RB, Roland CM, Ellis AR, Randall AM, Robertson CG. Effect of silica nanoparticles on the local segmental dynamics in poly(vinyl acetate). *Macromolecules* 2008;41:1289–1296.
244. Aso O, Eguiazabal JJ, Nazabal J. The influence of surface modification on the structure and properties of a nanosilica filled thermoplastic elastomer. *Compos Sci Technol* 2007;67:2854–2863.
245. Arrighi V, McEwen IJ, Qian H, Prieto MBS. The glass transition and interfacial layer in styrene-butadiene rubber containing silica nanofiller. *Polymer* 2003;44:6259–6266.
246. Gauthier C, Reynaud E, Vassoille R, Ladauce-Stelandre L. Analysis of the non-linear viscoelastic behaviour of silica filled styrene butadiene rubber. *Polymer* 2004;45:2761–2771.
247. Cassagnau P, Melis F. Non-linear viscoelastic behaviour and modulus recovery in silica filled polymers. *Polymer* 2003;44:6607–6615.
248. Wu TB, Ke YC. Melting, crystallization and optical behaviors of poly (ethylene terephthalate)-silica/polystyrene nanocomposite films. *Thin Solid Films* 2007;515:5220–5226.
249. Yan SF, Yin JB, Yang Y, Dai ZZ, Ma J, Chen XS. Surface-grafted silica linked with l-lactic acid oligomer: A novel nanofiller to improve the performance of biodegradable poly(l-lactide). *Polymer* 2007;48:1688–1694.
250. Kweon JO, Noh ST. Thermal, thermomechanical, and electrochemical characterization of the organic–inorganic hybrids poly(ethylene oxide) (PEO)–silica and PEO–silica–LiClO₄. *J Appl Polym Sci* 2001;81:2471–2479.
251. Mello NC, Bonagamba TJ, Panepucci H, Dahmouche K, Judeinstein P, Aegerter MA. NMR study of ion-conducting organic–inorganic nanocomposites poly(ethylene glycol)–silica–LiClO₄. *Macromolecules* 2000;33:1280–1288.
252. de Souza PH, Bianchi RF, Dahmouche K, Judeinstein P, Faria RM, Bonagamba TJ. Solid-state NMR, ionic conductivity, and thermal studies of lithium-doped siloxane–poly(propylene glycol) organic–inorganic nanocomposites. *Chem Mater* 2001;13:3685–3692.
253. Mark JE. *Science of Ceramic Chemical Processing*. New York: John Wiley & Sons; 1985. ; Chapter 17.
254. Erman B. *Structures and Properties of Rubberlike Networks*. New York: Oxford University Press; 1997. p 370.
255. Mark JE, Pan J. Reinforcement of polydimethylsiloxane networks by in-situ precipitation of silica: A new method for preparation of filled elastomers. *Makromol Chem Rapid Commun* 1982;3:681–685.
256. Kohjiya S, Ikeda Y. Reinforcement of general-purpose grade rubbers by silica generated in situ. *Rubber Chem Technol* 2000;73:534–550.
257. Hashim AS, Ikeda Y, Kohjiya S. Moisture cure and in-situ silica reinforcement of epoxidized natural rubber. *Polym Int* 1995;38:111–117.
258. Ikeda Y, Tanaka A, Kohjiya S. Effect of catalyst on in situ silica reinforcement of styrene–butadiene rubber vulcanizate by the sol–gel reaction of tetraethoxysilane. *J Mater Chem* 1997;7:455–458.
259. Ikeda Y, Tanaka A, Kohjiya S. Reinforcement of styrene–butadiene rubber vulcanizate by in situ silica prepared by the sol–gel reaction of tetraethoxysilane. *J Mater Chem* 1997;7:1497–1503.
260. Ikeda Y, Kohjiya S. In situ formed silica particles in rubber vulcanized by the sol–gel method. *Polymer* 1997;38:4417–4423.
261. Hashim AS, Azahari B, Ikeda Y, Kohjiya S. The effect of Bis(3-triethoxysilylpropyl) tetrasulfide on silica reinforcement of styrene-butadiene rubber. *Rubber Chem Technol* 1998;71:289–299.
262. Sun D, Zhang R, Liu Z, Huang Y, Wang Y, He J, Han B, Yang G. Polypropylene/silica nanocomposites prepared by in-situ sol–gel reaction with the aid of CO₂. *Macromolecules* 2005;38:5617–5624.
263. Jain S, Goossens H, Picchioni F, Magusin P, Mezari B, Van Duin M. Synthetic aspects and characterization of polypropylene–silica nanocomposites prepared via solid-state modification and sol–gel reactions. *Polymer* 2005;46:6666–6681.
264. Jain S, Goossens JGP, Van Duin M. Synthesis, characterization and properties of (vinyl triethoxy silane-grafted PP)/silica nanocomposites. *Macromol Symp* 2006;233:225–234.
265. Dou Q, Zhu X, Peter K, Demco DE, Moller M, Melian C. Preparation of polypropylene/silica composites by in situ sol–gel processing using hyperbranched polyethoxysiloxane. *J Sol-Gel Sci Technol* 2008;48:51–60.
266. Yu K, Zhao J, Guo Y, Ding X, Yanhua Liu HB, Wang Z. Sol–gel synthesis and hydrothermal processing of anatase nanocrystals from titanium n-butoxide. *Mater Lett* 2005;59:2515–2518.
267. Barlier V, Bounor-Legare V, Boiteux G, Davenas J, Slazak A, Rybak A, Jung J. Photogeneration and photovoltaic effect in poly(N-vinylcarbazole): TiO₂ bulk heterojunction elaborated by hydrolysis-condensation reactions of TiO₂ precursors. *Synth Met* 2009;159:508–512.
268. Barlier V, Bounor-Legare V, Boiteux G, Leonard D, Davenas J. TiO₂: polymer bulk-heterojunction thin films made from a miscible new carbazole based TiO₂ precursor with poly(Nvinylcarbazole) for enhanced charge transfer properties. *Mater Chem Phys* 2009;115:429–433.
269. Mazzocchi L, Scandola M, Pollicino A. Study of the organic-inorganic phase interactions in polyester-titanium hybrids. *Polymer* 2008;49:5215–5224.
270. Bahloul W, Bounorlegare V, David L, Cassagnau P. Morphology and viscoelasticity of PP/TiO₂ anocomposites prepared by in situ sol–gel method. *J Polym Sci B* 2010;48:1213–1222.
271. Peng-Cheng M, Naveed A, Siddiqui GM, Jang-Kyo K. Dispersion and functionalization of carbon nanotubes for polymer-based nanocomposites: A review. *Composites A* 2010;41:1345–1367.
272. Ma PC, Siddiqui NA, Cao SG, Tang BZ, Kim JK. Effect of dispersion techniques and functionalization of CNTs on the properties of CNT/epoxy composites. In: *Proceedings of the 2nd*

- international conference on advanced materials and structures*; 2008 October; Nanjing, China; 2008.
273. Dennis HR, Hunter DL, Chang D, Kim S, White JL, Cho JW, Paul DR. Effect of melt processing conditions on the extent of exfoliation organoclay-based nanocomposites. *Polymer* 2001;42:9513–9522.
274. Lertwimolnun W, Vergnes B. Effect of processing conditions on the formation of polypropylene/organoclay nanocomposites in a twin screw extruder. *Polym Eng Sci* 2006;46:314–323.
275. Lertwimolnun W, Vergnes B. Influence of compatibilizer and processing conditions on the dispersion of nanoclay in a polymer matrix. *Polymer* 2005;46:3462–3471.
276. Panacek A et al. Silver colloid nanoparticles: synthesis, characterization, and their antibacterial activity. *J Phys Chem B* 2006;110:248–253.
277. Nair LS, Laurencin CT. Silver nanoparticles: Synthesis and therapeutic applications. *J Biomed Nanotechnol* 2007;3:301–316.
278. Sun Y, Xia Y. Shape-controlled synthesis of gold and silver nanoparticles. *Science* 2002;298:2176–2179.